



Contents lists available at SciVerse ScienceDirect

Applied Mathematical Modelling

journal homepage: www.elsevier.com/locate/apm

3-D thermo-elastic solution for continuously graded isotropic and fiber-reinforced cylindrical shells resting on two-parameter elastic foundations

B. Sobhani Aragh^{a,*}, Aida Zeighami^b, Mohammad Rafiee^a, M.H. Yas^{c,d}, Magd Abdel wahab^e

^a Young Researchers Club, Arak Branch, Islamic Azad University, Arak, Iran

^b Department of Civil Engineering, Faculty of Engineering, Arak University, Arak, Iran

^c Mechanical Engineering Department, Razi University, Kermanshah, Iran

^d Center Excellence in Thermoelasticity, Amirkabir University of Technology, Tehran, Iran

^e Department of Mechanical Construction and Production, Faculty of Engineering, Ghent University, Gent, Belgium

ARTICLE INFO

Article history:

Received 11 October 2011

Received in revised form 4 December 2012

Accepted 9 January 2013

Available online xxx

Keywords:

Thermo-elastic

Generalized power-law distribution

Functionally graded material

Two-parameter elastic foundation

Orthotropic

ABSTRACT

This paper investigates the three-dimensional thermo-elastic deformation of cylindrical shells on two-parameter elastic foundations with continuously graded of volume fraction, subjected to thermal load. Suitable temperature and displacement functions that identically satisfy boundary conditions at the edges are used to reduce the equilibrium equations to a set of coupled ordinary differential equations with variable coefficients, which are solved by Generalized Differential Quadrature (GDQ) method. Results are presented for two-constituent isotropic and fiber-reinforced functionally graded cylindrical shells that have a smooth variation of volume fractions through the radial direction. Symmetric and asymmetric volume fraction profiles are presented in this paper. The fast rate of convergence of the method is demonstrated and comparison studies are carried out to establish its very high accuracy and versatility. Effects of stiffness of the foundation and variations of different parameters of generalized power-law distribution on steady-state responses of the functionally graded cylindrical shell resting on elastic foundation are discussed. In addition, the effects of the FGM configuration are studied by considering the mechanical entities of different FGM fiber-reinforced cylindrical shells resting on elastic foundation. Some results are presented for the first time and some important conclusions are drawn.

© 2013 Elsevier Inc. All rights reserved.

1. Introduction

Functionally graded materials (FGM) are those in which the volume fractions of two or more materials vary continuously as a function of position along certain direction(s) of the structure to achieve a required function. These novel materials were first introduced by a group of scientists in Sendai, Japan, in 1984 [1,2] and then were rapidly developed by other scientists. FGMs were initially designed as thermal barrier materials for aerospace structural applications and fusion reactors. These materials are now developed for general use as structural elements in extremely high-temperature environments. A list of different applications may be found in Ref. [3].

Topics such as; an in-depth discussion on various issues related to FGMs on modeling aspects to determine the magnitude of thermal stresses that develop due to large thermal loading, the optimal composition of the constituents or the

* Corresponding author. Tel.: +98 9364978879; fax: +98 8314274542.

E-mail address: behnamsobhani@yahoo.com (B. Sobhani Aragh).

composition profile that helps to decrease the magnitude of thermal stress, formation of cracks and eventually failure of the structural member by fracture and many others can be found in the article by Noda [4]. Fuchiyama and Noda [5] have developed a computer program to analyze the transient heat transfer and the transient thermal stresses in FGM components by the finite element method. Mirzavand and Eslami [6–8] investigated the effect of geometrical imperfections on the thermal and mechanical buckling of FGM cylindrical shells under several types of loadings for two different models of initial imperfections. Liew et al. [9] studied the analysis of the thermo-mechanical behavior of hollow circular cylinders of functionally graded material. Pelletier and Vel [10] presented generalized plane strain thermal stress of a functionally graded thick cylindrical shell subjected to thermal and mechanical loads. Katsuo et al. [11] studied the thermal stress distributions of laminated composite finite hollow cylinders restricted at both ends and at one end of the cylinders, using the thermo-elastic displacement potential and Michell's stress functions based on the axisymmetrical theory of elasticity. Bahtui and Eslami [12,13] studied the coupled thermo-elastic response of a functionally graded circular cylindrical shell. The coupled thermo-elastic and the energy equations were simultaneously solved for a functionally graded axisymmetric cylindrical shell subjected to thermal shock load. Jabbari et al. [14] presented the axisymmetric mechanical and thermal stresses in thick long FGM cylinders. They presented a direct method of solution of Navier equations, rather than potential functions method. The heat conduction and Navier equations were solved analytically, using the generalized Bessel function and complex form of Fourier integral. Obata and Noda [15] studied the steady thermal stresses in a hollow circular cylinder and a hollow sphere made of functionally graded materials. The aim of this research was to understand the effect of composition on stresses and to design the optimum FGM hollow circular cylinder and hollow sphere. Shahsiah and Eslami [16] studied thermal buckling of functionally graded cylindrical shell by using shell theories and finite element formulation. Reddy and Cheng [17] studied three-dimensional thermo-mechanical deformations of a simply supported Monel–zirconia functionally graded rectangular plate by using an asymptotic method. The locally effective material properties were estimated by the Mori–Tanaka scheme. Vel and Batra [18] presented an analytical solution for three dimensional thermo-mechanical deformations of a simply supported FGM plate subjected to time-dependent thermal loads on its top or bottom surfaces. Wang and Sudak [19] presented a general method for the analysis of a thermo-elastic multi-layered cylindrical panel made of an oblique pile of functionally graded layers having orthotropic material properties. Alibeigloo [20] studied three-dimensional thermo-elastic analysis of functionally graded rectangular plates with simply supported edges subjected to thermo-mechanical loads. Hosseini et al. [21] investigated the heat wave propagation and coupled thermo-elasticity without energy dissipation in functionally graded thick hollow cylinder based on Green–Naghdi theory. Yas and Sobhani Aragh [22] studied three-dimensional steady-state response of a functionally graded fiber reinforced cylindrical panel with graded fiber volume fractions. Results were compared with traditional discretely laminated composite panel. Three-dimensional analysis of thermal stresses in four-parameter continuous grading fiber reinforced cylindrical panels is carried out by Sobhani Aragh and Yas [23]. The influences of the power-law exponent, of the power-law distribution type and of the choice of the four parameters on the steady-state response of functionally graded cylindrical panels were considered. Recently, Jafari Mehrabadi and Sobhani Aragh [24] have studied thermoelastic response of a 2-D functionally graded open cylindrical shell with temperature-dependent material properties. The non-linear steady-state heat conduction equation was solved using an iterative solution procedure along with generalized differential quadrature method.

Cylindrical shells are usually laid on or placed in a soil medium as an elastic foundation, thus there is a great interest in analysis of the shells on elastic foundations. Paliwal et al. [25,26] investigated the free vibration of whole buried cylindrical shells with simply supported ends in contact with Winkler and Pasternak foundations using direct solution to the governing classical shell theory equations of motion. Yang et al. [27] investigated the behavior of whole buried pipelines subjected to sinusoidal seismic waves by the finite element method. Cai et al. [28] investigated free vibration of a cylindrical panel supported on Kerr foundation. Kerr model can be reduced to either a Pasternak model or a Winkler one by selecting certain values of foundation parameters. Gunawan et al. [29] examined the free vibrations of cylindrical shells partially buried in elastic foundations based on the finite element method. The shells are discretized into cylindrical finite elements and the distribution of the foundation in the circumferential direction is defined by Fourier series expansion. Farid et al. [30] studied three-dimensional temperature dependent free vibration analysis of functionally graded material curved panels resting on two-parameter elastic foundation subjected to thermal environment. Shah et al. [31] performed the vibrations of functionally graded cylindrical shells based on the Winkler and Pasternak foundations. A wave propagation approach is employed to solve the equations of motion of the shell involving the elastic foundation.

Despite the aforementioned extensive research on the analysis of cylindrical shells resting on elastic foundations, to the authors' best knowledge, there exists no study in the open literature for thermal analysis of functionally graded cylindrical shell resting on elastic foundations. The aim of this present study is to fill this apparent gap in this area by providing the thermal analysis of functionally graded isotropic and fiber-reinforced cylindrical shell resting on elastic foundations, subjected to thermal load with symmetric and asymmetric volume fractions. For the first time, comprehensive parametric investigations are presented for two-constituent isotropic and fiber-reinforced functionally graded cylindrical shells resting on elastic foundation with respect to different foundation stiffnesses and parameters of generalized power-law distribution. The effect of various Mori–Tanaka homogenization schemes on steady-state responses of the functionally graded cylindrical shells resting on elastic foundations is studied.

2. Problem formulation

2.1. Geometrical configuration

The geometric configuration of a FGM circular cylindrical shell is shown in Fig. 1. The coordinate system is located at the end plane of the cylindrical shell where the z -axis is directed along the longitudinal axis. The length, mean radius, uniform thickness and inner and outer radius of the shell are L , R , H , r_i and r_o , respectively. The inner surface is continuously in contact with an elastic medium that acts as an elastic foundation represented by the Winkler/Pasternak model with Winkler stiffness k_w and shearing layer stiffness k_g .

2.2. Material properties

We assume that the cylindrical shell is made of an isotropic/orthotropic material with material properties that are functions of the radial coordinate. Furthermore, we specifically consider two-phase isotropic or fiber-reinforced functionally graded cylindrical shells that have a smooth variation of volume fractions through the radial direction.

2.2.1. FGM isotropic shell

The effective material properties of the isotropic FGMs are determined in terms of the local volume fractions and material properties of the two isotropic phases by the Mori–Tanaka scheme. The Mori–Tanaka scheme [32,33] for estimating the effective moduli is applicable to regions of the graded microstructure which have a well-defined continuous matrix and a discontinuous particulate phase. It takes into account the interaction of the elastic fields among neighboring inclusions. It is assumed that the matrix phase, denoted by the subscript 1, is reinforced by spherical particles of a particulate phase, denoted by the subscript 2. In this notation, K_1 , G_1 , k_1 and α_1 are the bulk modulus, the shear modulus, the thermal conductivity and the thermal expansion coefficient, respectively, and V_1 is the volume fraction of the matrix phase. K_2 , G_2 , k_2 , α_2 and V_2 are the corresponding material properties and the volume fraction of the particulate phase. Note that $V_1 + V_2 = 1$, that the Lamé constant λ is related to the bulk and the shear moduli by $\lambda = K - 2G/3$, and that the stress–temperature modulus is related to the coefficient of thermal expansion by $\beta = (3\lambda + 2G)\alpha = 3K\alpha$. The following estimates for the effective local bulk modulus K and shear modulus G are useful for a random distribution of isotropic particles in an isotropic matrix:

$$\frac{K - K_1}{K_2 - K_1} = \frac{V_2}{1 + (1 - V_2)(K_2 - K_1)/(K_1 + (4/3)K_1)}, \quad (1)$$

$$\frac{G - G_1}{G_2 - G_1} = \frac{V_2}{1 + (1 - V_2)(G_2 - G_1)/(G_1 + f_1)}, \quad (2)$$

where $f_1 = G_1(9K_1 + 8G_1)/6(K_1 + 2G_1)$. The effective values of Young's modulus, E , and Poisson's ratio, ν , are found from:

$$E = \frac{9KG}{3K + G}, \quad \nu = \frac{3K - 2G}{2(3K + G)}. \quad (3)$$

The effective thermal conductivity k is given by Hatta and Taya [34]:

$$\frac{k - k_1}{k_2 - k_1} = \frac{V_2}{1 + (1 - V_2)(k_2 - k_1)/3k_1}. \quad (4)$$

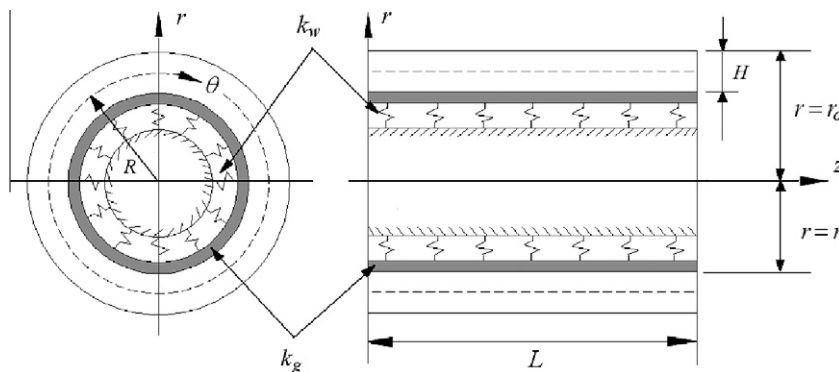


Fig. 1. Geometry of FGM cylindrical shell resting on elastic foundation.

And the coefficient of thermal expansion α is determined from the following relation [35]:

$$\frac{\alpha - \alpha_1}{\alpha_2 - \alpha_1} = \frac{1/K - 1/K_1}{1/K_2 - 1/K_1}. \quad (5)$$

2.2.2. FGM fiber-reinforced shell

The effective mechanical properties of the FGM fiber-reinforced cylindrical shell are obtained based on a micromechanical model as follows [36,37]:

$$E_{11} = V_f E_{11}^f + V_m E_{11}^m, \quad (6)$$

$$\frac{1}{E_{ii}} = \frac{V_f}{E_{ii}^f} + \frac{V_m}{E_{ii}^m} - V_f V_m \frac{v_f^2 E_{ii}^m / E_{ii}^f + v_m^2 E_{ii}^f / E_{ii}^m - 2 v_f v_m}{V_f E_{ii}^f + V_m E_{ii}^m}, \quad i = 2, 3, \quad (7)$$

$$\frac{1}{G_{ij}} = \frac{V_f}{G_{ij}^f} + \frac{V_m}{G_{ij}^m}, \quad ij = 12, 23 \text{ and } 13, \quad (8)$$

$$v_{ij} = V_f v_f^f + V_m v_m^m, \quad ij = 12, 23 \text{ and } 13, \quad (9)$$

$$\alpha_{11} = \frac{V_f E_{11}^f \alpha_{11}^f + V_m E_{11}^m \alpha_{11}^m}{V_f E_{11}^f + V_m E_{11}^m}, \quad (10)$$

$$\alpha_{ii} = (1 + v_f^f) V_f \alpha_{ii}^f + (1 + v_m^m) V_m \alpha_{ii}^m - v_{1i} \alpha_{11}, \quad i = 2, 3, \quad (11)$$

$$k_{11} = V_f k_{11}^f + V_m k_{11}^m, \quad (12)$$

$$\frac{1}{k_{ii}} = \frac{V_f}{k_{ii}^f} + \frac{V_m}{k_{ii}^m}, \quad i = 2, 3, \quad (13)$$

where E_{ii} , G_{ij} , v_{ij} , α_{ii} , k_{ii} , V_f and V_m are elasticity modulus, shear modulus, Poisson's ratio, thermal expansion coefficient, thermal conductivity, fiber volume fraction and matrix volume fraction, respectively.

For FGM isotropic/fiber-reinforced cylindrical shell, the volume fraction of the matrix phase is given by the power-law-type function:

$$V_1 = V^- + (V^+ - V^-) V_{fm}, \quad (14)$$

where V^- and V^+ , which have values that range from 0 to 1, denote the volume fractions of the matrix phase of two different materials. In this paper, V_{fm} follows simple four-parameter power-law distributions [22,38]:

$$V_{fm} = \left(1 - a \left(\frac{1}{2} - \frac{r-R}{H} \right) + b \left(\frac{1}{2} - \frac{r-R}{H} \right)^c \right)^p, \quad (15)$$

where the volume fraction index p ($0 \leq p \leq \infty$) and the parameters a , b , c dictate the material variation profile through the FGM cylindrical shell thickness.

For instance, by assuming $V^- = 1$ and $V^+ = 0.25$, some material profiles through the FGM cylindrical shell thickness can be illustrated in Figs. 2 and 3. In Fig. 2 the classical volume fraction profile are presented as special cases of the general distribution laws (15) by setting $a = 1$ and $b = 0$. With another choice of the parameters a , b , c , it is possible to obtain symmetric and asymmetric volume fraction profiles as shown in Fig. 3. In Fig. 3a, by setting $a = 1$, $b = 1$, and $c = 2$ the volume fraction of the matrix phase distribution present the same profiles by varying the volume fraction index p and are symmetric respect to the reference surface $\eta = 0$ of the cylindrical shell ($\eta = (r - R)/H$). Fig. 3b illustrates asymmetric profiles obtained by setting $a = 1$, $b = 1$ and $c = 5$. As shown from figures, we have the same constituent at the inner and outer surface but, unlike the previous case (Fig. 3a), profiles are not symmetric with respect to the reference surface $\eta = 0$ of the shell. In Fig. 4, matrix phase profile for the different values of parameters c is shown, by considering $p = 1$. In Fig. 4, matrix phase volume fraction profile along radial direction became asymmetric with increasing parameter c , also matrix phase volume fraction on the inner and outer surfaces of shell for the different values of parameter c is identical ($V_1 = V^+$).

2.3. Theoretical formulations

The thermo-elastic constitutive relations, which relates the stresses to the strains are as follows [39]:

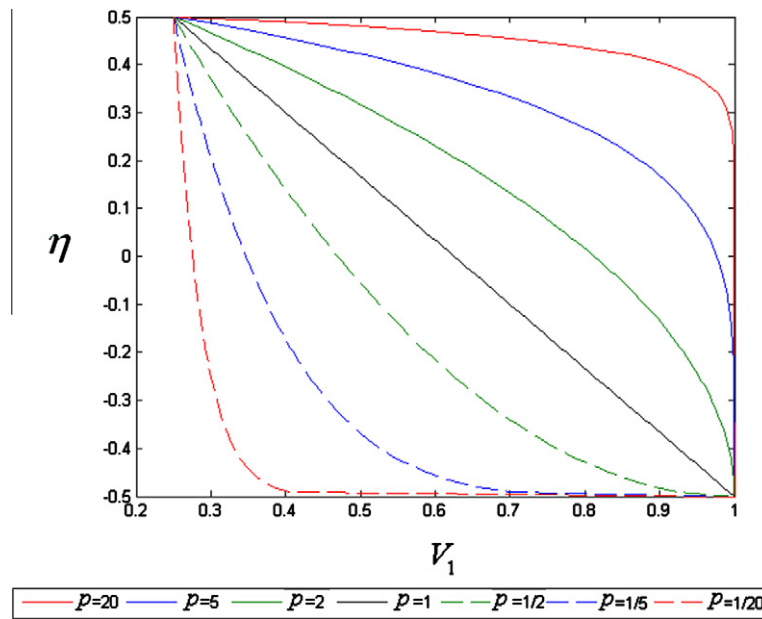


Fig. 2. Variations of the volume fractions of the matrix phase V_1 through the thickness for different values of the power-law index p ($a = 1, b = 0$).

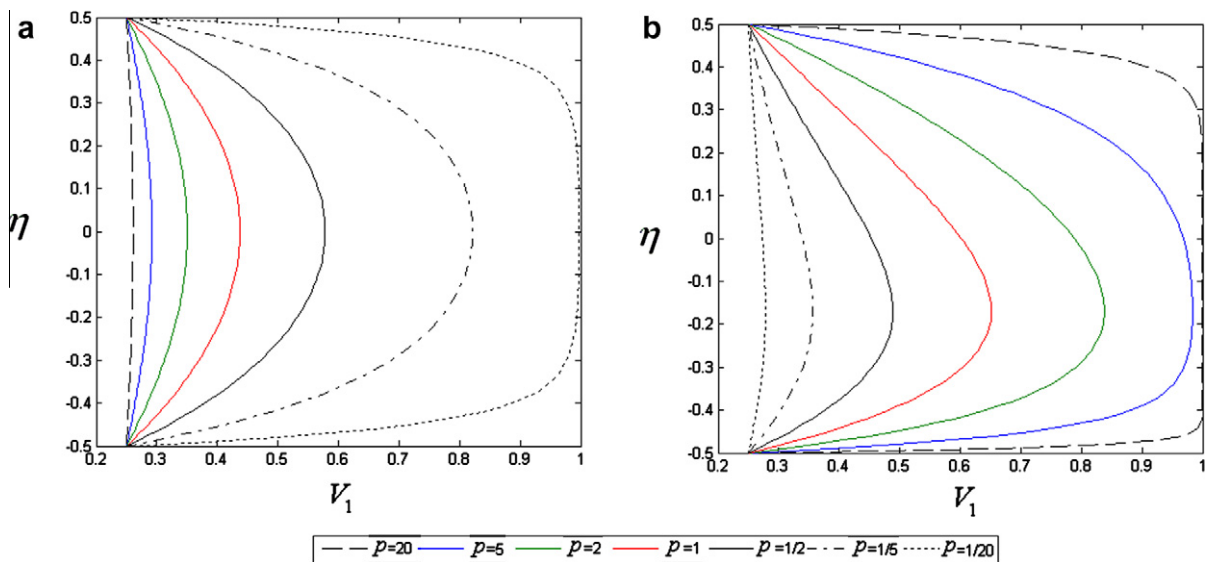


Fig. 3. Variations of the volume fractions of the matrix phase V_1 through the thickness for different values of the power-law index p (a: $a = 1, b = 1, c = 2$; b: $a = 1, b = 1, c = 5$).

$$\begin{Bmatrix} \sigma_z \\ \sigma_\theta \\ \sigma_r \\ \tau_{r\theta} \\ \tau_{zr} \\ \tau_{z\theta} \end{Bmatrix} = \begin{bmatrix} \bar{C}_{11} & \bar{C}_{12} & \bar{C}_{13} & 0 & 0 & 0 \\ \bar{C}_{12} & \bar{C}_{22} & \bar{C}_{23} & 0 & 0 & 0 \\ \bar{C}_{13} & \bar{C}_{23} & \bar{C}_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & \bar{C}_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & \bar{C}_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & \bar{C}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_z \\ \varepsilon_\theta \\ \varepsilon_r \\ \gamma_{r\theta} \\ \gamma_{zr} \\ \gamma_{z\theta} \end{Bmatrix} - \begin{bmatrix} \beta_{11} \\ \beta_{22} \\ \beta_{33} \\ 0 \\ 0 \\ 0 \end{bmatrix} T, \quad (16)$$

where $\bar{C}_{ij}$ are elastic stiffnesses and β_{ii} is related to the coefficients of thermal expansion α_{ii} as follows [10]:

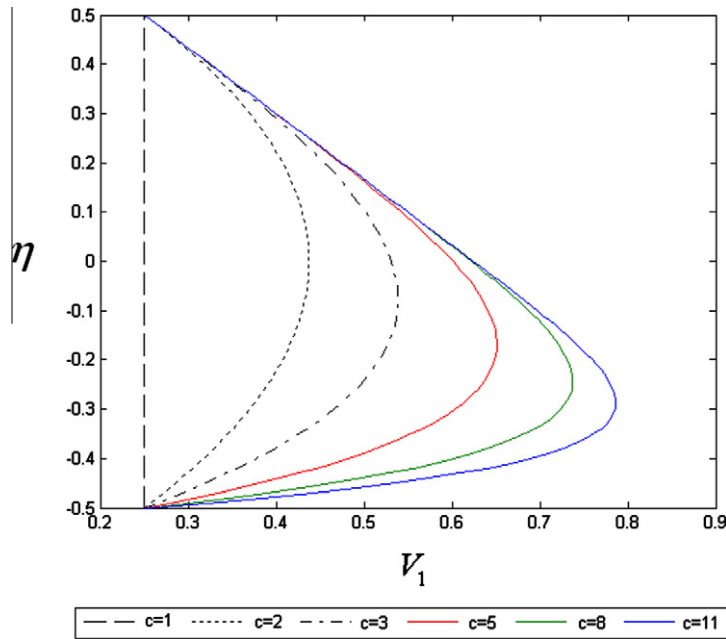


Fig. 4. Variations of the volume fractions of the matrix phase V_1 through the thickness ($p = 1$, $a = 1$, $b = 1$).

$$\begin{Bmatrix} \beta_{11} \\ \beta_{22} \\ \beta_{33} \end{Bmatrix} = \begin{Bmatrix} \bar{C}_{11}\alpha_{11} + \bar{C}_{12}\alpha_{22} + \bar{C}_{13}\alpha_{33} \\ \bar{C}_{12}\alpha_{11} + \bar{C}_{22}\alpha_{22} + \bar{C}_{23}\alpha_{33} \\ \bar{C}_{13}\alpha_{11} + \bar{C}_{23}\alpha_{22} + \bar{C}_{33}\alpha_{33} \end{Bmatrix}. \quad (17)$$

In the absence of body forces, the equilibrium equations are as follows [40]:

$$\begin{aligned} \frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{z\theta}}{r \partial \theta} + \frac{\partial \tau_{rz}}{\partial r} + \frac{\tau_{rz}}{r} &= 0, \\ \frac{\partial \tau_{z\theta}}{\partial z} + \frac{\partial \sigma_\theta}{r \partial \theta} + \frac{\partial \tau_{r\theta}}{\partial r} + \frac{2\tau_{r\theta}}{r} &= 0, \\ \frac{\partial \tau_{rz}}{\partial z} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\partial \sigma_r}{\partial r} + \frac{\sigma_r - \sigma_\theta}{r} &= 0. \end{aligned} \quad (18)$$

The three-dimensional steady-state heat conduction equation, expressed in terms of the local coordinates, is:

$$\frac{\partial q_r}{\partial r} + \frac{1}{r} \frac{\partial q_\theta}{\partial \theta} + \frac{\partial q_z}{\partial z} + \frac{q_r}{r} = 0, \quad (19)$$

where q_θ , q_r and q_z are the components of the heat flux vector. Fourier's law of heat conduction, which relates the heat flux to the temperature gradient, is:

$$\begin{Bmatrix} q_z \\ q_\theta \\ q_r \end{Bmatrix} = - \begin{Bmatrix} k_{11} & 0 & 0 \\ 0 & k_{22} & 0 \\ 0 & 0 & k_{33} \end{Bmatrix} \begin{Bmatrix} \frac{\partial T}{\partial z} \\ \frac{1}{r} \frac{\partial T}{\partial \theta} \\ \frac{\partial T}{\partial r} \end{Bmatrix}, \quad (20)$$

where T is the change in temperature of a material particle from that in the stress-free reference configuration and k_{ii} are the thermal conductivities. Since the shell is graded in the radial direction, the elastic stiffnesses ($\bar{C}_{ij}$), coefficients of thermal expansion (α_{ii}) and thermal conductivities (k_{ii}) are functions of the radial coordinate. The infinitesimal strain tensor is related to the displacements as follows:

$$\begin{aligned} \varepsilon_z &= \frac{\partial u_z}{\partial z}, \quad \varepsilon_\theta = \frac{u_r}{r} + \frac{\partial u_\theta}{r \partial \theta}, \quad \varepsilon_r = \frac{\partial u_r}{\partial r}, \quad \varepsilon_{z\theta} = \frac{1}{2} \left(\frac{\partial u_\theta}{\partial z} + \frac{\partial u_z}{r \partial \theta} \right), \quad \varepsilon_{r\theta} = \frac{1}{2} \left(\frac{-u_\theta}{r} + \frac{\partial u_\theta}{\partial r} + \frac{\partial u_r}{r \partial \theta} \right), \\ &= \frac{1}{2} \left(\frac{\partial u_z}{\partial r} + \frac{\partial u_r}{\partial z} \right), \end{aligned} \quad (21)$$

where u_r , u_θ and u_z are radial, circumferential and axial displacement components, respectively. Upon substitution Eq. (21) into (16) and then into (18), also Eq. (20) into (19), the following equilibrium equations are obtained:

$$\begin{bmatrix} L_{1r} & L_{1\theta} & L_{1z} & L_{1T} \\ L_{2r} & L_{2\theta} & L_{2z} & L_{2T} \\ L_{3r} & L_{3\theta} & L_{3z} & L_{3T} \\ 0 & 0 & 0 & L_{4T} \end{bmatrix} \begin{Bmatrix} u_r \\ u_\theta \\ u_z \\ T \end{Bmatrix} = \{0\}. \quad (22)$$

Expressions for the coefficients L_{ij} are shown in Appendix A.

Boundary conditions imposed at the edges ($z = 0, L$) of the cylindrical shell resting on elastic foundation are given as follows:

$$u_r = 0, \quad \sigma_\theta = 0, \quad \sigma_z = 0, \quad T = 0 \quad \text{at} \quad z = 0, L \quad (23)$$

Moreover the boundary conditions on the inner ($r = r_i$) and outer ($r = r_o$) surfaces of the cylindrical shell resting on elastic foundation are as follows:

$$\begin{aligned} \tau_{rz} = \tau_{r\theta} = 0, \quad \sigma_r = -k_w u_r + k_g \Delta u_r, \quad T = 0 \quad \text{at} \quad r = r_i, \\ \sigma_r = \tau_{rz} = \tau_{r\theta} = 0, \quad T = T_o \quad \text{at} \quad r = r_o, \end{aligned} \quad (24)$$

where $\Delta = \partial^2 / \partial z^2 + (1/r^2) \partial^2 / \partial \theta^2$.

A semi-inverse solution to the heat conduction problem is considered by assuming the following form for the temperature field:

$$T(r, \theta, z) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} T(r) \sin(m\theta) \sin\left(\frac{n\pi}{L} z\right). \quad (25)$$

The assumed form of the temperature field identically satisfies the homogeneous boundary conditions (23) and (24) for the temperature at the edges. In addition the following solutions that satisfy the boundary conditions at the edges ($z = 0, L$) is assumed:

$$\begin{aligned} u_r(r, \theta, z) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} U_r(r) \sin(m\theta) \sin\left(\frac{n\pi}{L} z\right), \\ u_\theta(r, \theta, z) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} U_\theta(r) \cos(m\theta) \sin\left(\frac{n\pi}{L} z\right), \\ u_z(r, \theta, z) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} U_z(r) \sin(m\theta) \cos\left(\frac{n\pi}{L} z\right), \end{aligned} \quad (26)$$

where m and n are number of circumferential and axial waves of sinusoidal distributed thermal load.

Upon substituting Eqs. (25) and (26) into the governing equation (22), the coupled partial differential equations are reduced to a set of ordinary differential relationships as follows:

$$\begin{bmatrix} J_{1r} & J_{1\theta} & J_{1z} & J_{1T} \\ J_{2r} & J_{2\theta} & J_{2z} & J_{2T} \\ J_{3r} & J_{3\theta} & J_{3z} & J_{3T} \\ 0 & 0 & 0 & J_{4T} \end{bmatrix} \begin{Bmatrix} U_r \\ U_\theta \\ U_z \\ T \end{Bmatrix} = \{0\}, \quad (27)$$

where coefficients J_{ij} are given in Appendix B.

3. Solution of governing equations

In this paper, the GDQ approach is used to solve the governing equations of FGM cylindrical shell resting on elastic foundation. GDQ approach was developed by Shu and co-workers [41,42] based on the (DQ) technique [43]. It approximates the spatial derivative of a function at a given grid point as a weighted linear sum of all the functional values at all grid points in the whole domain. The computation of weighting coefficients by GDQ is based on the analysis of a high order polynomial approximation and the analysis of a linear vector space. The weighting coefficients of the first-order derivative are calculated by a simple algebraic formulation, and the weighting coefficient of the second- and higher-order derivatives are given by a recurrence relationship. The details of GDQ method can be found in Refs [41,42].

In GDQ method, the n th order partial derivative of a continuous function $f(x, z)$ with respect to x at a given point x_i can be approximated as a linear sum of weighted function values at all of the discrete points in the domain of x , i.e.

$$\frac{\partial f^{n(x_i, z)}}{\partial x^n} = \sum_{k=1}^N c_{ik}^n f(x_{ik}, z), \quad (i = 1, 2, \dots, N, n = 1, 2, \dots, N-1), \quad (28)$$

where N is the number of sampling points, and c_{ij}^n is the x_i dependent weight coefficients. In order to determine the weighting coefficients c_{ij}^n , the Lagrange interpolation basic functions are used as the test functions and explicit formula for computing these weighting coefficients can be obtained as [41,42]:

$$c_{ij}^{(1)} = \frac{M^{(1)}(x_i)}{(x_i - x_j)M^{(1)}(x_i)} \quad i, j = 1, 2, \dots, N, \quad i \neq j, \quad (29)$$

where

$$M^{(1)}(x_i) = \prod_{j=1, j \neq i}^N (x_i - x_j). \quad (30)$$

For first order derivative; i.e. $n = 1$ and for higher-order derivatives, one can use the following relations iteratively:

$$c_{ij}^{(n)} = n \left(c_{ii}^{(n-1)} c_{ij}^{(1)} - \frac{c_{ij}^{(n-1)}}{(x_i - x_j)} \right) \quad i, j = 1, 2, \dots, N, \quad i \neq j, \quad n = 2, 3, \dots, N-1, \quad (31)$$

$$c_{ii}^{(n)} = - \sum_{j=1, j \neq i}^N c_{ij}^{(n)} \quad i = 1, 2, \dots, N, \quad n = 1, 2, \dots, N-1. \quad (32)$$

A simple and natural choice of the grid distribution is the uniform grid spacing rule. However, it was found that non-uniform grid spacing yields results with better accuracy [44]. Hence, in this work, the Chebyshev–Gauss–Lobatto quadrature points are used, that is [42]:

$$r_i = \frac{1}{2} \left(1 - \cos \left(\frac{i-1}{N-1} \pi \right) \right), \quad i = 1, 2, \dots, N. \quad (33)$$

Eq. (28) being applied to Eq. (27) at each domain grid point r_i with $i = 2, \dots, N-1$ the discretized equations take the following forms:

$$\begin{aligned} & - \left[(\bar{C}_{55})_i \left(\frac{n\pi}{L} \right)^2 + (\bar{C}_{44})_i \frac{1}{r^2} m^2 - \left(\frac{\partial \bar{C}_{23}}{\partial r} \right)_i \frac{1}{r} + (\bar{C}_{22})_i \frac{1}{r^2} \right] U_{ri} + \left[(\bar{C}_{44})_i \frac{1}{r^2} + (\bar{C}_{22})_i \frac{1}{r^2} - \left(\frac{\partial \bar{C}_{23}}{\partial r} \right)_i \frac{1}{r} \right] m U_{\theta i} \\ & + \left[- \left(\frac{\partial \bar{C}_{13}}{\partial r} \right)_i - (\bar{C}_{13})_i \frac{1}{r} + (\bar{C}_{12})_i \frac{1}{r} \right] \left(\frac{n\pi}{L} \right) U_{zi} + \left[- \left(\frac{\partial \beta_{33}}{\partial r} \right)_i - \frac{1}{r} (\beta_{33})_i + (\beta_{22})_i \frac{1}{r} \right] T_i \\ & + \sum_{k=1}^N \left(\left[\left(\frac{\partial \bar{C}_{33}}{\partial r} \right)_i + (\bar{C}_{33})_i \frac{1}{r} \right] c_{ik}^{(1)} + (\bar{C}_{33})_i c_{ik}^{(2)} \right) U_{rk} - [(\bar{C}_{44})_i + (\bar{C}_{23})_i] \frac{1}{r} m \sum_{k=1}^N c_{ik}^{(1)} U_{\theta k} - \beta_{33} \sum_{k=1}^N c_{ik}^{(1)} T_k \\ & - [(\bar{C}_{13})_i + (\bar{C}_{55})_i] \left(\frac{n\pi}{L} \right) \sum_{k=1}^N c_{ik}^{(1)} U_{zk} = 0, \\ & \left[(\bar{C}_{22})_i \frac{1}{r^2} + (\bar{C}_{44})_i \frac{1}{r^2} + \left(\frac{\partial \bar{C}_{44}}{\partial r} \right)_i \frac{1}{r} \right] m U_{ri} - \left[(\bar{C}_{66})_i \left(\frac{n\pi}{L} \right)^2 + (\bar{C}_{22})_i \frac{1}{r^2} m^2 + (\bar{C}_{44})_i \frac{1}{r^2} + \left(\frac{\partial \bar{C}_{44}}{\partial r} \right)_i \frac{1}{r} \right] U_{\theta i} \\ & - \left[(\bar{C}_{66})_i \frac{1}{r} + (\bar{C}_{12})_i \frac{1}{r} \right] \left(\frac{n\pi}{L} \right) m U_{zi} - \frac{1}{r} (\beta_{22})_i m T_i + [(\bar{C}_{23})_i + (\bar{C}_{44})_i] \frac{1}{r} m \sum_{k=1}^N c_{ik}^{(1)} U_{rk} \\ & + \sum_{k=1}^N \left(\left[\left(\frac{\partial \bar{C}_{44}}{\partial r} \right)_i + (\bar{C}_{44})_i \frac{1}{r} \right] c_{ik}^{(1)} + (\bar{C}_{44})_i c_{ik}^{(2)} \right) U_{\theta k} = 0, \quad (35) \end{aligned}$$

$$\begin{aligned} & \left[(\bar{C}_{12})_i \frac{1}{r} + (\bar{C}_{55})_i \frac{1}{r} + \left(\frac{\partial \bar{C}_{55}}{\partial r} \right)_i \right] \left(\frac{n\pi}{L} \right) U_{ri} - [(\bar{C}_{12})_i + (\bar{C}_{66})_i] \frac{1}{r} \left(\frac{n\pi}{L} \right) m U_{\theta i} - \left[(\bar{C}_{11})_i \left(\frac{n\pi}{L} \right)^2 + (\bar{C}_{66})_i \frac{1}{r^2} m^2 \right] U_{zi} \\ & - (\beta_{11})_i \left(\frac{n\pi}{L} \right) T_i + [(\bar{C}_{13})_i + (\bar{C}_{55})_i] \left(\frac{n\pi}{L} \right) \sum_{k=1}^N c_{ik}^{(1)} U_{rk} + \sum_{k=1}^N \left(\left[\left(\frac{\partial \bar{C}_{55}}{\partial r} \right)_i + (\bar{C}_{55})_i \frac{1}{r} \right] c_{ik}^{(1)} + (\bar{C}_{55})_i c_{ik}^{(2)} \right) U_{zk} = 0, \quad (36) \end{aligned}$$

$$-\left[(k_{22})_i \frac{1}{r^2} m^2 + (k_{11})_i \left(\frac{n\pi}{L}\right)^2\right] T_i + \sum_{k=1}^N \left[\left[(k_{33})_i \frac{1}{r} + \left(\frac{\partial k_{33}}{\partial r}\right)_{i,j} \right] c_{ik}^{(1)} + (k_{33})_i c_{ik}^{(2)} \right] T_k = 0, \quad (37)$$

In the above equation $c_{ik}^{(1)}$ and $c_{ik}^{(2)}$ are the weighting coefficients of the first and second order derivatives. The discretized forms of the boundary conditions on the inner and outer surfaces of the cylindrical shell, shown in Eq. (34), can be expressed as follows. On the inner surface ($r = r_i$):

$$\begin{aligned} &-(\bar{C}_{13})_1 \left(\frac{n\pi}{L}\right) U_{z1} + (\bar{C}_{23})_1 \frac{1}{r_i} U_{r1} - (\bar{C}_{23})_1 \frac{1}{r_i} m U_{\theta 1} + (\bar{C}_{33})_1 \sum_{k=1}^N c_{1k}^{(1)} U_{rk} - (\beta_{33})_1 T_1 - \left\{ k_w + k_g \left(\frac{n\pi}{L}\right)^2 + k_g m^2 (1/r_i^2) \right\} U_{r1} = 0, \\ &-(\bar{C}_{44})_1 \frac{1}{r_i} U_{\theta 1} + (\bar{C}_{44})_1 \sum_{k=1}^N c_{1k}^{(1)} U_{\theta k} + (\bar{C}_{44})_1 \frac{1}{r_i} m U_{r1} = 0, \\ &(\bar{C}_{55})_1 \sum_{k=1}^N c_{1k}^{(1)} U_{zk} + (\bar{C}_{55})_1 \left(\frac{n\pi}{L}\right) U_{r1} = 0, \\ &T_1 = 0, \end{aligned} \quad (38)$$

On the outer surface ($r = r_o$):

$$\begin{aligned} &-(\bar{C}_{13})_N \left(\frac{n\pi}{L}\right) U_{zN} + (\bar{C}_{23})_N \frac{1}{r_o} U_{rN} - (\bar{C}_{23})_N \frac{1}{r_o} m U_{\theta N} + (\bar{C}_{33})_N \sum_{k=1}^N c_{Nk}^{(1)} U_{rk} - (\beta_{33})_1 T_N = 0, \\ &-(\bar{C}_{44})_N \frac{1}{r_o} U_{\theta N} + (\bar{C}_{44})_N \sum_{k=1}^N c_{Nk}^{(1)} U_{\theta k} + (\bar{C}_{44})_N \frac{1}{r_o} m U_{rN} = 0, \\ &(\bar{C}_{55})_N \sum_{k=1}^N c_{Nk}^{(1)} U_{zk} + (\bar{C}_{55})_N \left(\frac{n\pi}{L}\right) U_{rN} = 0, \\ &T_N = T_o, \end{aligned} \quad (39)$$

To perform the eigenvalue system of equations, the degrees of freedom are separated into the domain and the boundary degrees of freedom as [41,45]:

$$d = \begin{Bmatrix} U_r \\ U_\theta \\ U_z \\ T \end{Bmatrix}_{\text{domain}}, \quad b = \begin{Bmatrix} U_r \\ U_\theta \\ U_z \\ T \end{Bmatrix}_{\text{boundary}} \quad (40)$$

Using Eq. (40), the discretized form of the equilibrium equations in matrix form can be rearranged as

$$\mathbf{S}_{db} \mathbf{b} + \mathbf{S}_{dd} \mathbf{d} = 0, \quad (41)$$

where $\mathbf{S}_{db}$ and $\mathbf{S}_{db}$ are stiffness matrices. In a similar manner, the discretized form of the boundary conditions becomes:

$$\mathbf{S}_{bb} \mathbf{b} + \mathbf{S}_{bd} \mathbf{d} = \mathbf{q}, \quad (42)$$

where $\mathbf{S}_{bb}$ and $\mathbf{S}_{bd}$ are stiffness matrices and $\mathbf{q}$ is thermal load. In the above equations, the elements of stiffness matrices are obtained based on the definition of vectors of domain and boundary degrees of freedom from the differential quadrature discretized form of the equilibrium equations and the boundary conditions. Using Eq. (42) to eliminate the boundary degrees of freedom $\mathbf{b}$ from Eq. (41), one obtains:

$$\mathbf{S} \mathbf{d} = \mathbf{S}_{db} \mathbf{S}_{bb}^{-1} \mathbf{q}, \quad (43)$$

where

$$\mathbf{S} = \mathbf{S}_{db} \mathbf{S}_{bb}^{-1} \mathbf{S}_{bd} - \mathbf{S}_{dd}.$$

The above system of equations can be solved to find the displacement and temperature fields of the FGM cylindrical shell resting on elastic foundation.

4. Results and discussion

4.1. Validation and comparison studies

To validate the approach, the present results are obtained for 90° single layer orthotropic cylindrical shell under internal static load and compared with similar results by Varadan and Bhaskar [46]. The mechanical properties of the considered material are as follows:

$$\frac{E_L}{E_T} = 25, \quad \frac{G_{TT}}{E_T} = 0.2, \quad \frac{G_{LT}}{E_T} = 0.5, \quad \nu_{LT} = \nu_{TT} = 0.25. \quad (44)$$

And non- dimensional parameters are:

$$(\bar{\sigma}_z, \bar{\sigma}_\theta) = \frac{10(\sigma_z, \sigma_\theta)}{qS^2}, \quad \bar{\sigma}_r = \frac{\sigma_r}{q}, \quad \bar{\tau}_{r\theta} = \frac{10\tau_{r\theta}}{qS}, \quad S = \frac{R}{H}. \quad (45)$$

As it is observed from Table 1 there is good agreement between the results.

For further validation, the results for an FGM isotropic cylindrical shell are studied under uniform thermal load and compared with published results by Ruhi et al. [47]. The analysis is conducted using aluminum as the outer surface metal and zirconia as the inner surface ceramic. The following material properties are used in computing the numerical results:

$$E_{out} = E_{Al} = 70 \text{ GPa}, \quad \alpha_{out} = \alpha_{Al} = 23 \times 10^{-6} \text{ K}^{-1},$$

$$E_{in} = E_{Cer} = 151 \text{ GPa}, \quad \alpha_{in} = \alpha_{Cer} = 10 \times 10^{-6} \text{ K}^{-1},$$

$$\nu = 0.3.$$

And $H = 20 \text{ mm}$, $r_i = 2.5H$, $L = 10r_i$.

The material property gradation through the thickness in Ref. [47] can be modeled by setting $a = 1$ and $b = 0$ in Eq. (15). Fig. 5 shows the variation of the radial stress ($\bar{\sigma}_r = \sigma_r / \alpha_{Cer} \Delta T$) through the thickness ($\eta = (r - R)/H$) for different power law exponent p of classical volume fraction profile ($a = 1, b = 0$) at the middle length of the cylinder due to the uniform thermal loading. It is noticed the present results agree very well with the literature.

4.2. FGM cylindrical shell resting on elastic foundation

In this section, we characterize the response of isotropic/orthotropic cylindrical shells resting on elastic foundation with continuously graded of volume fraction under thermal load. The FGM isotropic cylindrical shells considered in the examples are assumed to be composed of aluminum and silicon carbide. We also study FGM orthotropic cylindrical shells made of continuous tungsten reinforcement fibers in a copper matrix. These material combinations have found widespread use in high performance applications [48]. The relevant material properties for the constituent materials are listed in Table 2. We assume the material properties are independent of temperature. Non dimensional temperature, displacements, stresses are defined as follows:

Table 1

Comparison of the maximum stresses for 90° single layer orthotropic cylindrical shell ($N = 13$, $L/R = 4$, $p = 0$).

S		$\bar{\sigma}_\theta(\eta = \pm 0.5)$	$\bar{\sigma}_z(\eta = \pm 0.5)$	$\bar{\sigma}_r(\eta = 0)$	$\bar{\sigma}_{r\theta}(\eta = 0)$
2	Present	-14.8825	-0.78390	-0.37409	-2.05563
		5.163241	0.13317		
	[46]	-14.883	-0.7839	-0.37	-2.056
10	Present	-4.50902	-0.06558	-1.372343	-3.66901
		4.05089	0.06632		
	[46]	-4.509	-0.0656	-1.37	-3.669
50	Present	-3.97856	-0.00857	-5.38282	-3.91932
		3.90184	0.08447		
	[46]	-3.979	-0.0086	-5.38	-3.919
100	Present	-3.87638	0.02878	-10.12841	-3.85904
		3.84309	0.11897		
	[46]	-3.876	0.0288	-10.13	-3.859
		3.843	0.119		

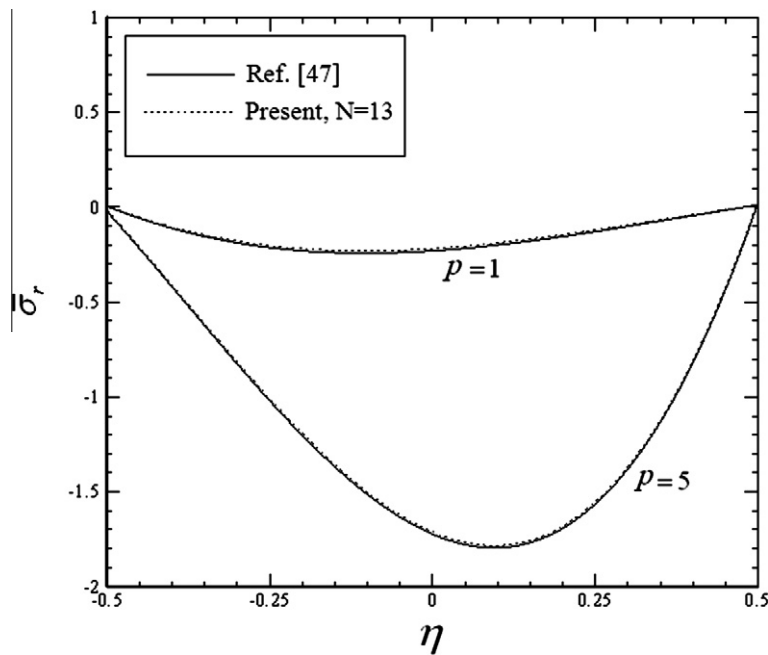


Fig. 5. Comparison of the non-dimensional radial stress of FGM cylindrical shell under uniform thermal load for different values of power law exponent p ($a = 1$, $b = 0$).

Table 2

Mechanical properties of aluminum, silicon carbide, copper and tungsten.

	Al	SiC	Cu	W
$E(\text{GPa})$	70	427	115	400
ν	0.3	0.17	0.31	0.28
$\alpha (10^{-6}/\text{K})$	23.4	4.3	17	4.4
$k (\text{W/mK})$	233	65	391	163

$$\bar{T} = \frac{T}{T_0}, \quad U = \frac{10U}{\alpha^i T_0}, \quad \bar{\sigma} = \frac{\sigma}{\alpha^i E^i T_0} \quad (46)$$

where α^i and E^i are material properties of matrix phase. The non-dimensional forms of the elastic foundation coefficients are defined as $K_w = k_w E^i H / [R^2 (1 - \nu^j)]$ and $K_g = k_g E^i H / (1 - \nu^j)$ in which ν^j is the Poisson's ratio of matrix phase. The cylindrical shell has geometrical parameters as: $L = 2 \text{ m}$, $R = 0.5 \text{ m}$.

A convergence study of the numerical results for FGM isotropic cylindrical shell resting on elastic foundation is shown in Fig. 6, by setting $a = 1$, $b = 1$, $c = 2$. The results are obtained by using the Mori–Tanaka scheme with the metal as the matrix phase and the ceramic as the particulate phase. As observed fast rate of convergence of the method is evident and it can also be seen that for the considered system, the formulation is stable while increasing the number of points.

4.2.1. FGM isotropic cylindrical shell

We use the label M–Tm to denote the Mori–Tanaka method with metal (Al) as the matrix phase, M–Tc to denote the Mori–Tanaka method with ceramic (SiC) as the matrix phase. For example, in the M–Tm method, $p_1 = p_m$ and $p_2 = p_c$, where p stands for either the volume fraction or a material property. The variations of Young's modulus and the shear modulus corresponding to the two homogenization schemes for $c = 2$ and $c = 5$ are shown in Fig. 7a and b.

Fig. 8 shows the effect of Winkler elastic coefficient on the non-dimensional radial stress $\bar{\sigma}_r(\eta = 0)$ of FGM isotropic cylindrical shell for different values of shearing layer elastic coefficients. In Fig. 8, symmetric volume fraction profile is considered by setting $a = 1$, $b = 1$, $c = 2$. The effective material properties are obtained using the M–Tm scheme. It is also observed for the large values of Winkler elastic coefficient, the shearing layer elastic coefficient has less effect and the results become independent of it; in other word the non-dimensional radial stress converge with increasing Winkler foundation stiffness.

The influence of shearing layer elastic coefficient on the non-dimensional radial stress $\bar{\sigma}_r(\eta = 0)$ is shown in Fig. 9. It results the variations of Winkler elastic coefficient has little effect on the non-dimensional radial stress at different values of shearing layer elastic coefficient. From Figs. 8 and 9, it can be inferred that the most effective ranges of Winkler foundation

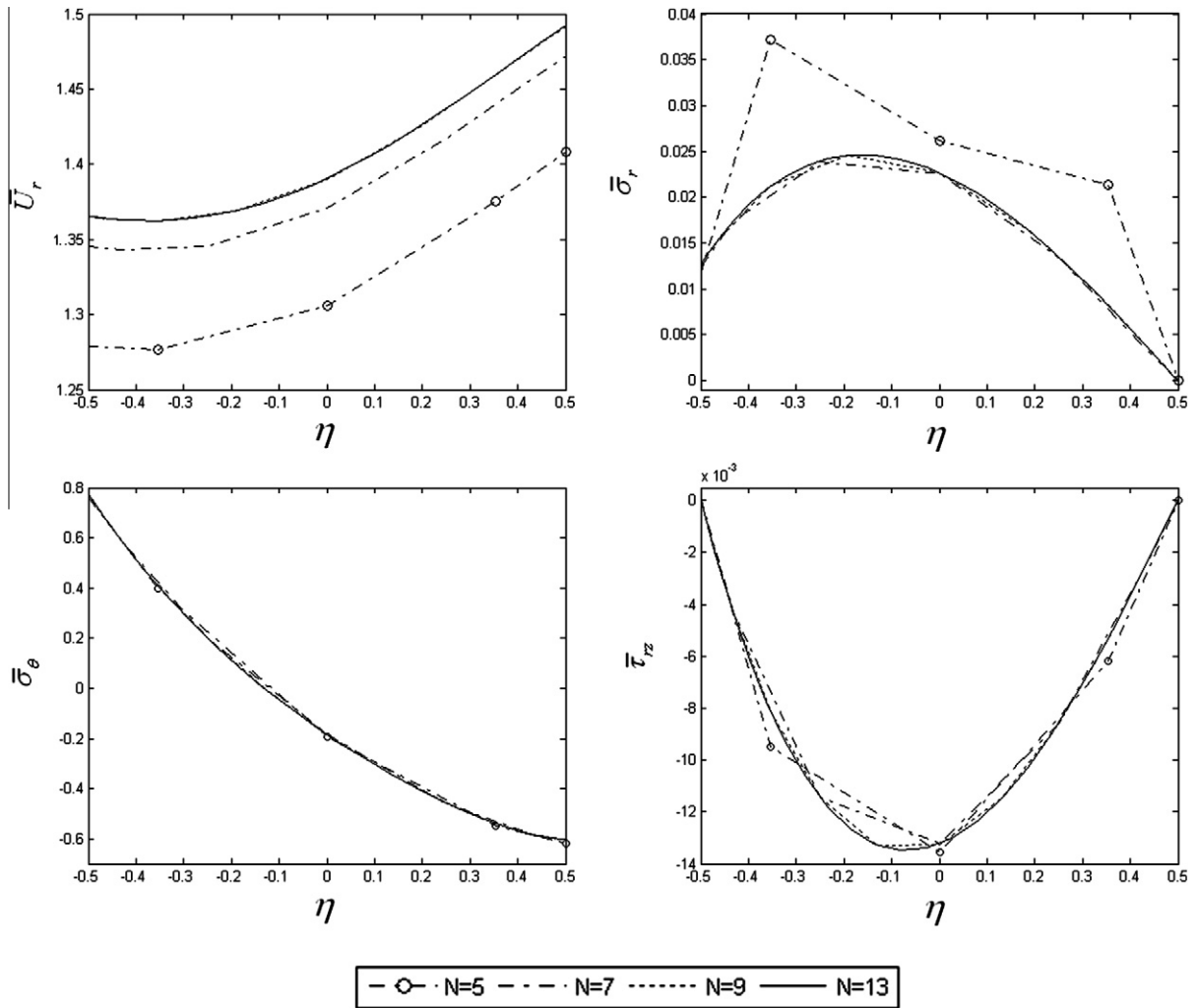


Fig. 6. Convergence of the non-dimensional radial displacement and radial, circumferential and transverse shear stresses through the thickness of FGM isotropic cylindrical shell resting on elastic foundation ($p = 1$, $a = 1$, $b = 1$, $c = 2$, $S = 10$, $K_w = 1e2$, $K_g = 1e-2$).

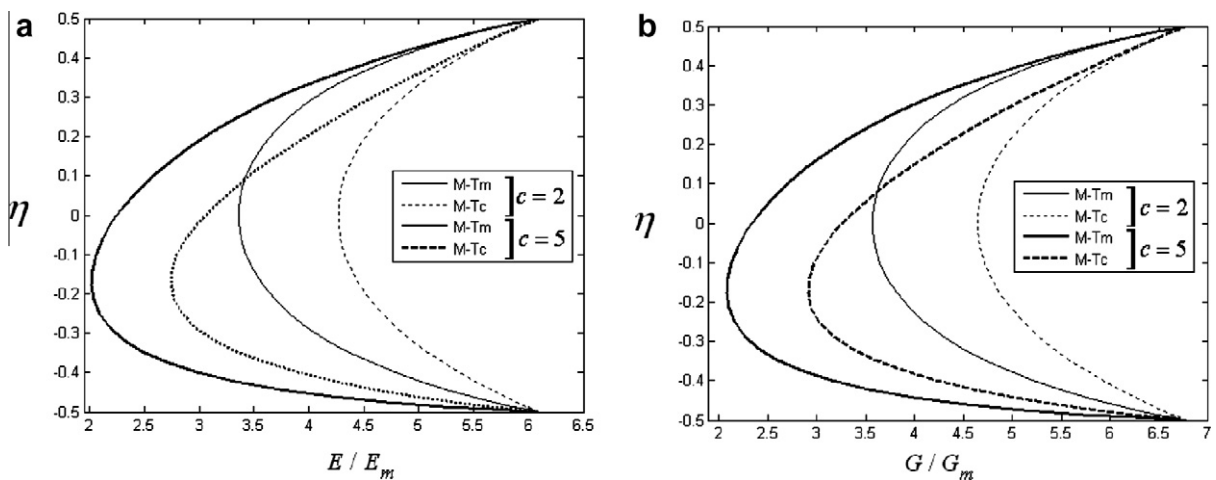


Fig. 7. Through-the-thickness distribution of (a) Young's modulus and (b) the shear modulus, corresponding to the two homogenization schemes ($a = b = 1$).

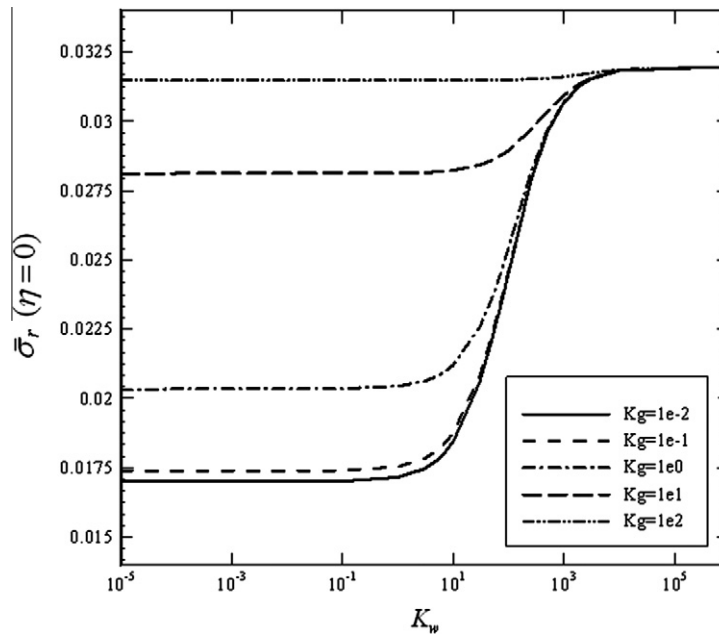


Fig. 8. Variation of the non-dimensional radial stress $\bar{\sigma}_r(\eta=0)$ of FGM isotropic cylindrical shell vs. Winkler elastic coefficient for different shearing layer elastic coefficients using the M-Tm homogenization scheme ($p=1$, $a=1$, $b=1$, $c=2$, $S=10$).

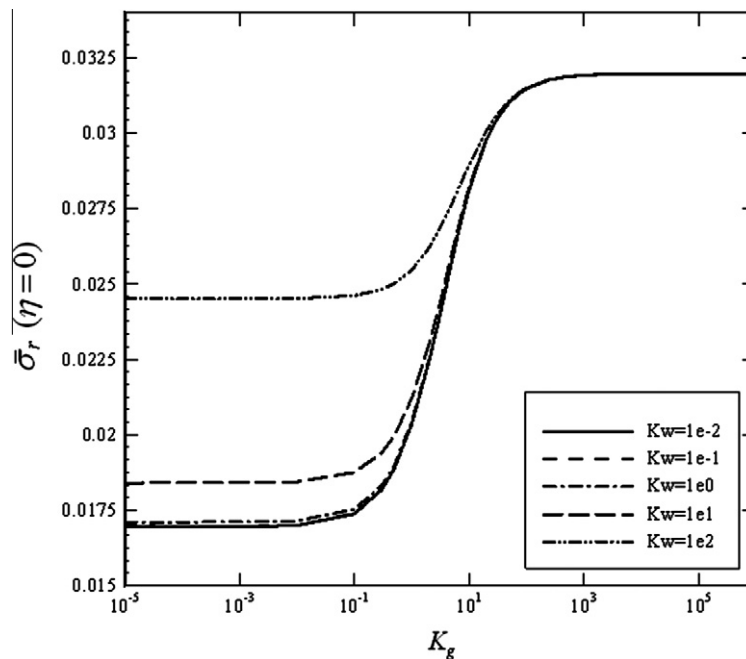


Fig. 9. Variation of the non-dimensional radial stress $\bar{\sigma}_r(\eta=0)$ of FGM isotropic cylindrical shell vs. shearing layer elastic coefficients for different Winkler elastic coefficients using the M-Tm homogenization scheme ($p=1$, $a=1$, $b=1$, $c=2$, $S=10$).

stiffness and shearing layer elastic coefficient in increasing the non-dimensional radial stress are from 10^2 to 10^4 and 1 to 10^2 , respectively.

In Table 3, the maximum value of the non-dimensional radial and transverse shear stresses, the non-dimensional axial stresses at the inner surface and the non-dimensional radial displacement at the outer surface for different values of Winkler elastic coefficient and mid radius to thickness ratios, S , are listed. It is noticed that the maximum values of the non-dimensional radial and transverse shear stresses decrease with increasing the S ratio. Also, it can be seen that increasing the S ratio, the sensitivity of the non-dimensional mechanical entities with respect to elastic foundations coefficients reduces.

Table 3Variation of the mechanical entities of FGM isotropic cylindrical shell against Winkler elastic coefficient and S ratios ($p = 1$, $a = 1$, $b = 1$, $c = 2$, $K_g = 1e - 2$).

S	K_w	$\sigma_{r_{\max}}$	$\sigma_{rz_{\max}}$	$\sigma_z(\eta = -0.5)$	$\bar{U}_r(\eta = 0.5)$
10	1e-1	0.017104	-0.012457	0.888876	3.025706
	1e0	0.017237	-0.012477	0.884957	2.996514
	1e1	0.018438	-0.012654	0.849721	2.734093
	1e2	0.024601	-0.013489	0.682939	1.491956
	1e3	0.030662	-0.014241	0.532745	0.373368
	1e4	0.031866	-0.014381	0.504759	0.164934
	1e5	0.031997	-0.014396	0.501716	0.142277
20	1e-1	0.008904	-0.006689	0.880054	2.875256
	1e0	0.009155	-0.006708	0.831798	2.763738
	1e1	0.010917	-0.006838	0.774628	1.997362
	1e2	0.014505	-0.007081	0.567283	0.570377
	1e3	0.015673	-0.007156	0.503047	0.128289
	1e4	0.015815	-0.007165	0.495249	0.074624
	1e5	0.015829	-0.007166	0.494453	0.069143
50	1e-1	0.003821	-0.002781	0.876815	2.577009
	1e0	0.004264	-0.002795	0.801652	2.082141
	1e1	0.005567	-0.002834	0.595661	0.725911
	1e2	0.006202	-0.002851	0.503486	0.119039
	1e3	0.006288	-0.002853	0.490961	0.036577
	1e4	0.006297	-0.002853	0.489663	0.028034
	1e5	0.006298	-0.002854	0.489533	0.027176
100	1e-1	0.002174	-0.001409	0.791022	1.981549
	1e0	0.002575	-0.001415	0.661106	1.138332
	1e1	0.003032	-0.001422	0.520594	0.226342
	1e2	0.003131	-0.001424	0.491400	0.036855
	1e3	0.003141	-0.001424	0.488163	0.015844
	1e4	0.003143	-0.001424	0.487835	0.013720
	1e5	0.003143	-0.001424	0.487803	0.013508

Fig. 10 shows through-the-thickness variation of non-dimensional radial and axial stresses with respect to different values of parameter p , where the material properties have been obtained using the M-Tm and M-Tc, by setting $a = 1$, $b = 1$, $c = 2$ (symmetric profile). As can be seen from Fig. 10a, the peak of the non-dimensional radial stress increases and tends to the inner surface with increasing parameter p . As can be seen from Fig. 10b, axial stress increases through the thickness for $p = 0$ but by increasing the parameter p , it decreases slowly towards the middle surface and then increases rapidly. The peak of the axial stress occurs at the outer surface. The M-Tm scheme predicts larger mechanical entities of FGM isotropic cylindrical shell than that predicted by the M-Tc.

Fig. 11 illustrates variation of the temperature field along the thickness of FGM isotropic cylindrical shell for symmetric and asymmetric profiles using M-Tm scheme for different values of parameter p . As observed in Fig. 11a and b, the distri-

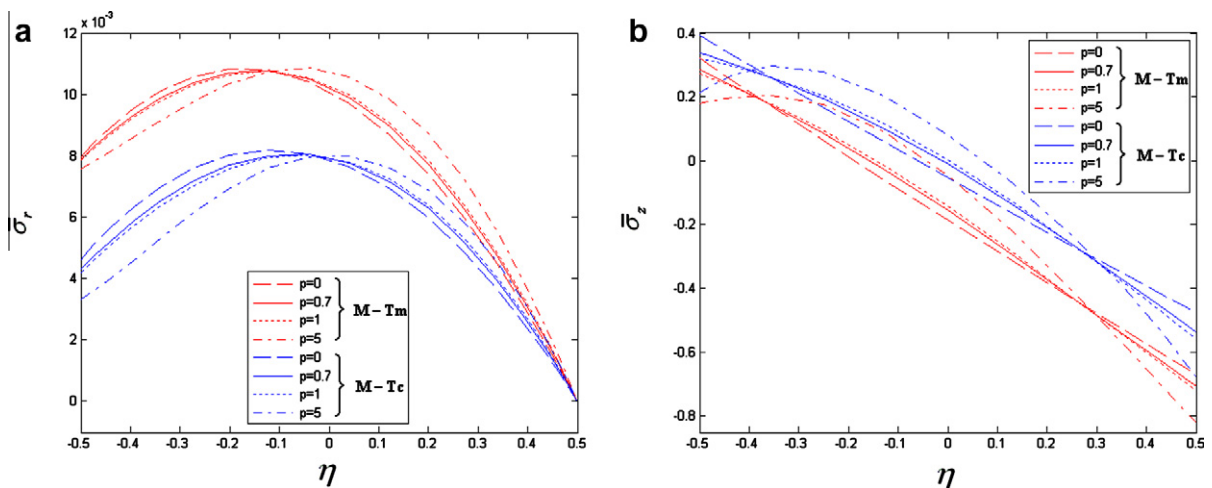


Fig. 10. Through-the-thickness variation of the mechanical entities of FGM isotropic cylindrical shell with respect to different values of p using various Mori-Tanaka homogenization schemes ($a = 1$, $b = 1$, $c = 2$, $S = 20$, $K_w = 1e2$, $K_g = 1e - 2$).

bution of the temperature field is almost linear for $p = 0$ and with increasing parameter p , temperature field becomes non-linear for both profiles. Nonlinearity rate is more pronounced near the outer surface of the shell for asymmetric profile while the distribution of the temperature for symmetric profile is symmetric with respect to mid-surface for different values of parameter p .

4.2.2. FGM fiber-reinforced orthotropic cylindrical shell

The orthotropic cylindrical shell consists of continuous tungsten reinforcement fibers in a copper matrix (W/Cu). The effective mechanical properties of the fiber reinforced cylindrical shell are obtained based on a micromechanical model presented in Eqs. (6)–(13). Distribution of radial, transverse shear, circumferential stresses and temperature field through the thickness of FGM fiber-reinforced cylindrical shell resting on elastic foundation for various parameter c ($1 \leq c \leq 11$) are presented in Fig. 12a–d, by setting $a = 1$, $b = 1$, $p = 1$. For $c = 1$, the cylindrical shell is composed of 25% copper and 75% tungsten and for $c = 2$, the shape of the profile volume fraction of matrix phase along radial direction is symmetric. In Figs. 12a–d, matrix phase volume fraction profile along radial direction becomes asymmetric with increasing parameter c ($c > 2$). Also matrix phase volume fraction on the inner and outer surface of cylindrical shell for the different values of parameter c is the same ($V_m = 0.25$). The distribution of the temperature field is almost linear for $c = 1$ and becomes nonlinear with increasing parameter c . As shown in Fig. 12b and d, the peak of the non-dimensional radial and transverse shear stresses for symmetric profile is larger than for asymmetric profile. Also the peaks of the non-dimensional radial and transverse shear stresses decrease and tend to the inner surface with increasing parameter c .

Distribution of the radial displacement as well as radial, circumferential, axial and transverse shear stresses through the thickness of FGM fiber-reinforced cylindrical shell for different Winkler elastic coefficient are presented in Fig. 13a–d, by setting $a = 1$, $b = 1$, $c = 2$ (symmetric profile). Fig. 13a and d show that the non-dimensional radial and transverse shear stresses of FGM fiber-reinforced cylindrical shell increase with increasing Winkler elastic coefficient. On the contrary, the circumferential and axial stresses through the thickness decrease with increasing Winkler elastic coefficient. As observed in Fig. 13a–d, the mechanical entities of FGM fiber-reinforced cylindrical shell converge with increasing Winkler foundation stiffness.

The effects of the FGM configuration are studied by considering the mechanical entities of four FGM fiber-reinforced cylindrical shells resting on elastic foundation. Volume fraction profiles along the thickness of the different types of FGM fiber-reinforced cylindrical shells are illustrated in Fig. 14. Symmetric and asymmetric volume fraction profiles are considered by setting $a = 1$, $b = 1$, $c = 2$, $p = 1$ and $a = 1$, $b = 1$, $c = 5$, $p = 1$, respectively. With specifying $a = 1$, $b = 0$, $p = 1$ in Eq. (15), it is possible to obtain Classic I profile. Moreover, Classic II profile can be acquired by modifying Eq. (15) as follows:

$$V_{fm} = \left(1 - a \left(\frac{1}{2} + \frac{r-R}{H} \right) + b \left(\frac{1}{2} + \frac{r-R}{H} \right)^c \right)^p \quad (47)$$

By setting $a = 1$, $b = 0$, $p = 1$ in Eq. (47) Classic II profile can be obtained. Fig. 15a–e show the variation of the non-dimensional radial, transverse shear, axial stresses and temperature field through the thickness of the different types of FGM fiber-reinforced cylindrical shells resting on elastic foundation. As observed in Fig. 15a–e, the mechanical entities at a point in classic II FGM cylindrical shell are greater than that at the corresponding point in the other types of FGM cylindrical shell. Fig. 15a–e show that the variation of the mechanical entities through the thickness for symmetric FGM cylindrical shells

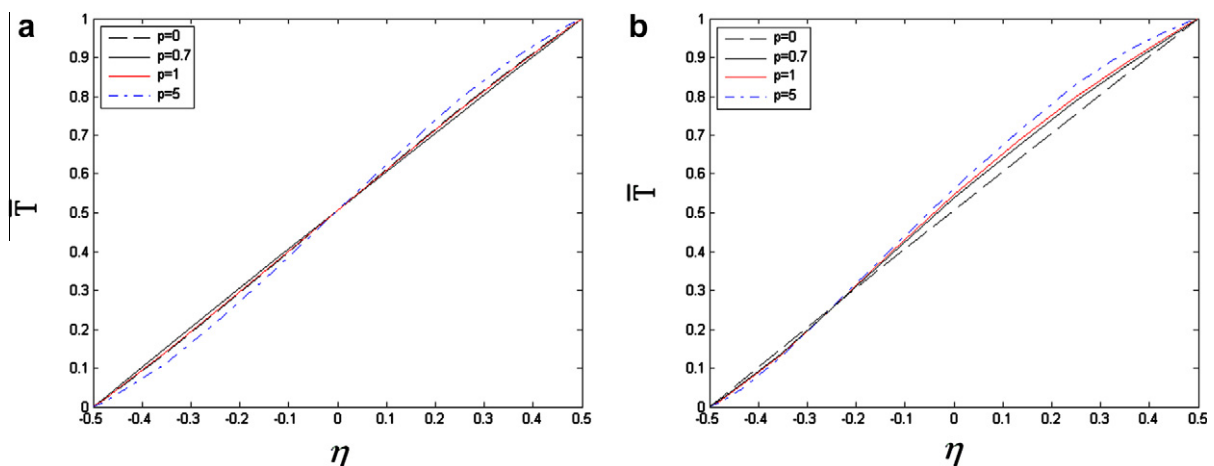


Fig. 11. Variation of the temperature field along the thickness of FGM isotropic cylindrical shell for (a) Symmetric ($c = 2$) and (b) Asymmetric ($c = 5$) profiles using M–Tm scheme ($a = 1$, $b = 1$, $S = 20$, $K_w = 1e1$, $K_g = 1e-2$).

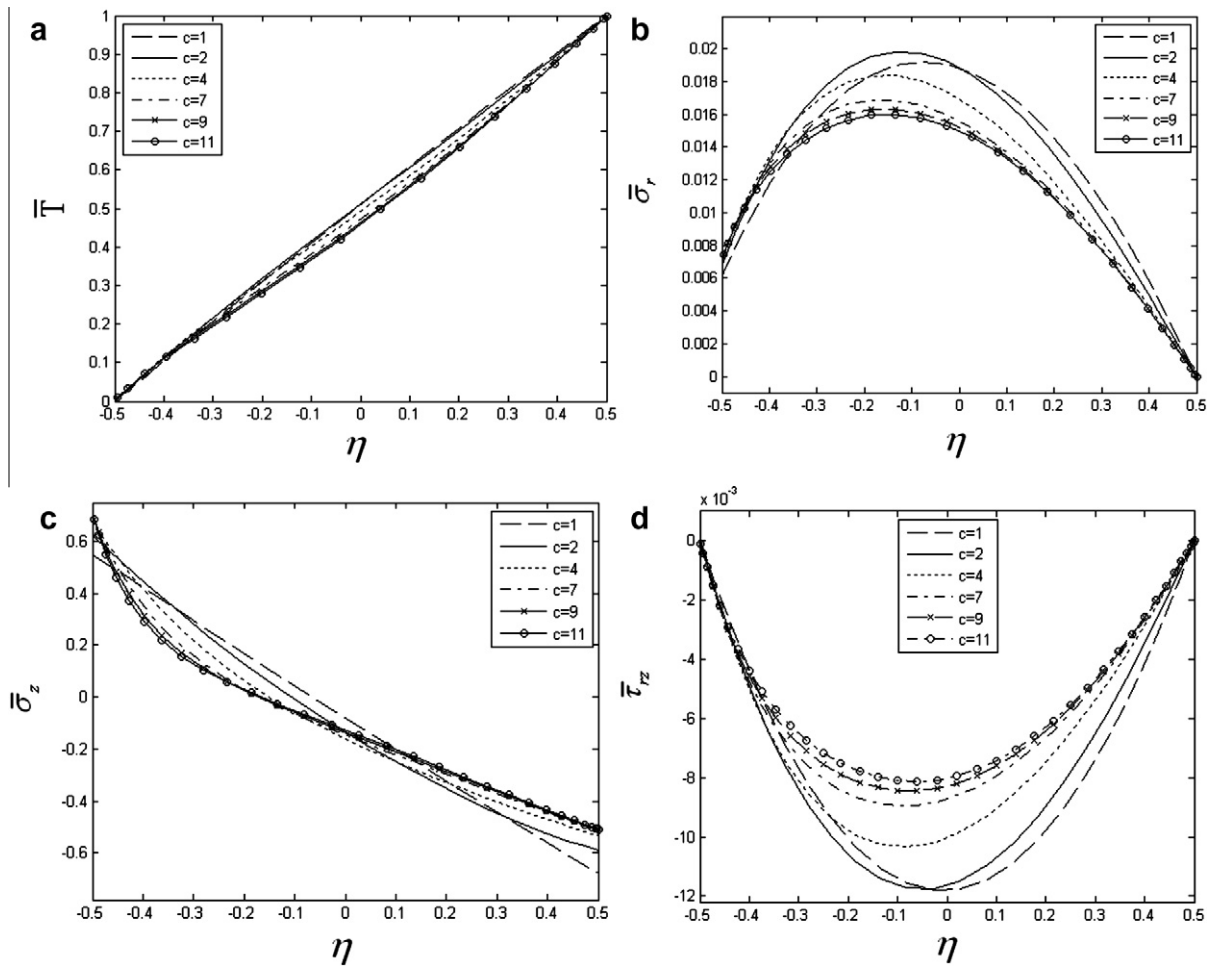


Fig. 12. Through-the-thickness variation of the temperature field and radial, axial and transverse shear stresses along the thickness of FGM fiber-reinforced cylindrical shell resting on elastic foundation for various values of the parameter c ($p = 1$, $a = 1$, $b = 1$, $S = 10$, $K_w = 1e2$, $K_g = 1e - 1$).

is almost the same as that for asymmetric FGM cylindrical shells. As noticed in Fig. 15b and d, the distribution of radial and transverse shear stresses along the thickness in classic I FGM cylindrical shell is smaller than that of the other types of FGM cylindrical shells.

5. Conclusions

In this research study, three-dimensional thermo-elastic deformation of cylindrical shells on two-parameter elastic foundations with continuously graded volume fraction, subjected to thermal load is considered. The effective material properties at a point in the cylindrical shell are inferred from the volume fractions and material properties of the constituent phases of FGM isotropic and fiber-reinforced using the Mori–Tanaka method and the micromechanical model, respectively. The constituent volume fractions and hence the effective material properties are assumed to vary in the radial direction. The generalized differential quadrature method (GDQM) is used to solve the 3-D thermo elasticity equations. For the first time, comprehensive parametric investigations are presented for two-constituent isotropic and fiber-reinforced functionally graded cylindrical shells resting on elastic foundation with respect to different foundation stiffnesses and parameters of generalized power-law distribution. From the study, some conclusions can be made;

- Results show that the M–Tm scheme predicts larger mechanical entities of FGM isotropic cylindrical shell than that predicted by the M–Tc.
- It is established that elastic foundations affects significantly the thermal behavior of functionally graded isotropic and fiber-reinforced cylindrical shells.
- The variations of Winkler elastic coefficient has little effect on the non-dimensional radial stress of functionally graded isotropic and fiber-reinforced cylindrical shell at different values of shearing layer elastic coefficient.

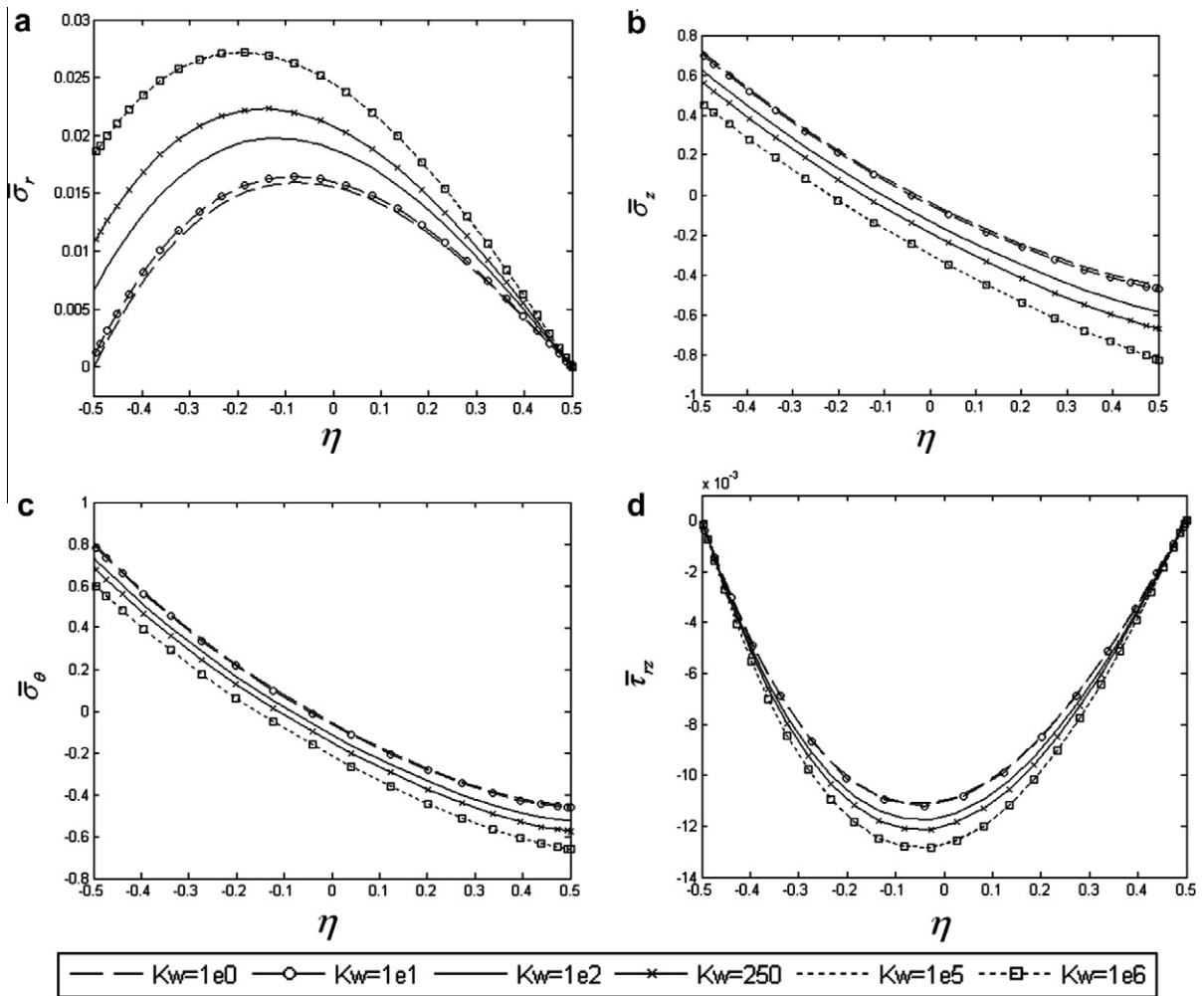


Fig. 13. Distribution of the non-dimensional radial, axial, circumferential and transverse shear stresses through the thickness of FGM fiber-reinforced cylindrical shell with respect to different Winkler elastic coefficients ($p = 1$, $a = 1$, $b = 1$, $c = 2$, $S = 10$, $K_g = 1e-2$).

- Results show that for the large values of Winkler elastic coefficient, the shearing layer elastic coefficient has less effect on the non-dimensional radial stress.
- Results indicate that by increasing the radius to thickness ratio, the sensitivity of the non-dimensional mechanical entities of FGM isotropic cylindrical shell with respect to elastic foundations coefficients reduces.
- It is shown that the variation of the mechanical entities along the thickness for symmetric FGM cylindrical shells is almost the same as that for asymmetric FGM cylindrical shells. It is also shown that the mechanical entities at a point in classic II FGM cylindrical shell are greater than that at the corresponding point in the other types of FGM cylindrical shell.

Appendix A. Expressions for the coefficients L_{ij} in Eq. (22)

$$L_{1r} = \bar{C}_{55} \frac{\partial^2}{\partial z^2} + \frac{\partial \bar{C}_{33}}{\partial r} \frac{\partial}{\partial r} + \bar{C}_{33} \frac{\partial^2}{\partial r^2} + \bar{C}_{44} \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial \bar{C}_{23}}{\partial r} \frac{1}{r} + \bar{C}_{23} \frac{1}{r} \frac{\partial}{\partial r} - \bar{C}_{22} \frac{1}{r^2} + \bar{C}_{33} \frac{1}{r} \frac{\partial}{\partial r} - \bar{C}_{23} \frac{1}{r} \frac{\partial}{\partial r},$$

$$L_{1\theta} = -\bar{C}_{44} \frac{1}{r^2} \frac{\partial}{\partial \theta} + \bar{C}_{44} \frac{1}{r} \frac{\partial^2}{\partial r \partial \theta} + \frac{\partial \bar{C}_{23}}{\partial r} \frac{1}{r} \frac{\partial}{\partial \theta} + \bar{C}_{23} \frac{1}{r} \frac{\partial}{\partial r \partial \theta} - \bar{C}_{22} \frac{1}{r^2} \frac{\partial}{\partial \theta},$$

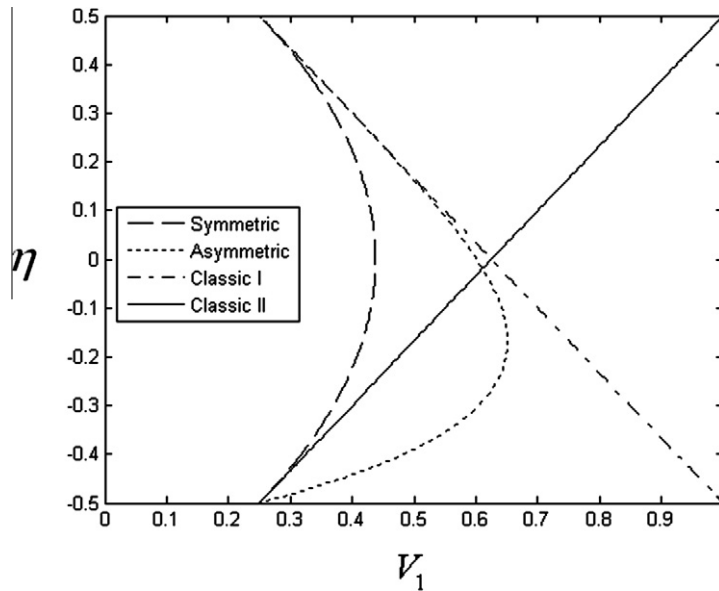


Fig. 14. Variation of the volume fractions of the matrix phase V_1 through the thickness for different types of FGM fiber-reinforced cylindrical shells ($p = 1$).

$$L_{1z} = \bar{C}_{13} \frac{1}{r} \frac{\partial}{\partial z} - \bar{C}_{12} \frac{1}{r} \frac{\partial}{\partial z} + \frac{\partial \bar{C}_{13}}{\partial r} \frac{\partial}{\partial z} + \bar{C}_{13} \frac{\partial}{\partial r \partial z} \bar{C}_{55} \frac{\partial^2}{\partial r \partial z},$$

$$L_{1T} = -\beta_{33} \frac{\partial}{\partial r} + \beta_{22} \frac{1}{r} - \frac{\partial \beta_{33}}{\partial r} - \beta_{33} \frac{1}{r},$$

$$L_{2r} = \bar{C}_{44} \frac{2}{r^2} \frac{\partial}{\partial \theta} + \bar{C}_{23} \frac{1}{r} \frac{\partial}{\partial r \partial \theta} + \frac{\partial \bar{C}_{44}}{\partial r} \frac{1}{r} \frac{\partial}{\partial \theta} - \bar{C}_{44} \frac{1}{r^2} \frac{\partial}{\partial \theta} + \bar{C}_{44} \frac{1}{r} \frac{\partial^2}{\partial r \partial \theta} + \bar{C}_{22} \frac{1}{r^2} \frac{\partial}{\partial \theta},$$

$$L_{2\theta} = \bar{C}_{66} \frac{\partial^2}{\partial z^2} - \frac{\partial \bar{C}_{44}}{\partial r} \frac{1}{r} + \frac{\partial \bar{C}_{44}}{\partial r} \frac{\partial}{\partial r} + \bar{C}_{44} \frac{\partial^2}{\partial r^2} + \bar{C}_{22} \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} - \bar{C}_{44} \frac{1}{r^2} + \bar{C}_{44} \frac{1}{r} \frac{\partial}{\partial r},$$

$$L_{2z} = \bar{C}_{66} \frac{1}{r} \frac{\partial}{\partial \theta \partial z} + \bar{C}_{12} \frac{1}{r} \frac{\partial}{\partial \theta \partial z},$$

$$L_{2T} = -\frac{1}{r} \beta_{22} \frac{\partial}{\partial \theta},$$

$$L_{3r} = \bar{C}_{12} \frac{1}{r} \frac{\partial}{\partial z} + \bar{C}_{13} \frac{\partial^2}{\partial r \partial z} + \frac{\partial \bar{C}_{55}}{\partial r} \frac{\partial}{\partial z} + \bar{C}_{55} \frac{\partial^2}{\partial r \partial z} + \bar{C}_{55} \frac{1}{r} \frac{\partial}{\partial z},$$

$$L_{3\theta} = \bar{C}_{12} \frac{1}{r} \frac{\partial^2}{\partial \theta \partial z} + \bar{C}_{66} \frac{1}{r} \frac{\partial^2}{\partial \theta \partial z},$$

$$L_{3z} = \bar{C}_{11} \frac{\partial}{\partial z^2} + \bar{C}_{66} \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial \bar{C}_{55}}{\partial r} \frac{\partial}{\partial r} + \bar{C}_{55} \frac{\partial^2}{\partial r^2} + \bar{C}_{55} \frac{1}{r} \frac{\partial}{\partial r},$$

$$L_{3T} = -\beta_1 \frac{\partial}{\partial z},$$

$$L_{4T} = \frac{\partial k_{33}}{\partial r} \frac{\partial}{\partial r} + k_{33} \frac{\partial^2}{\partial r^2} + k_{22} \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + k_{11} \frac{\partial^2}{\partial z^2} + k_{33} \frac{1}{r} \frac{\partial}{\partial r},$$

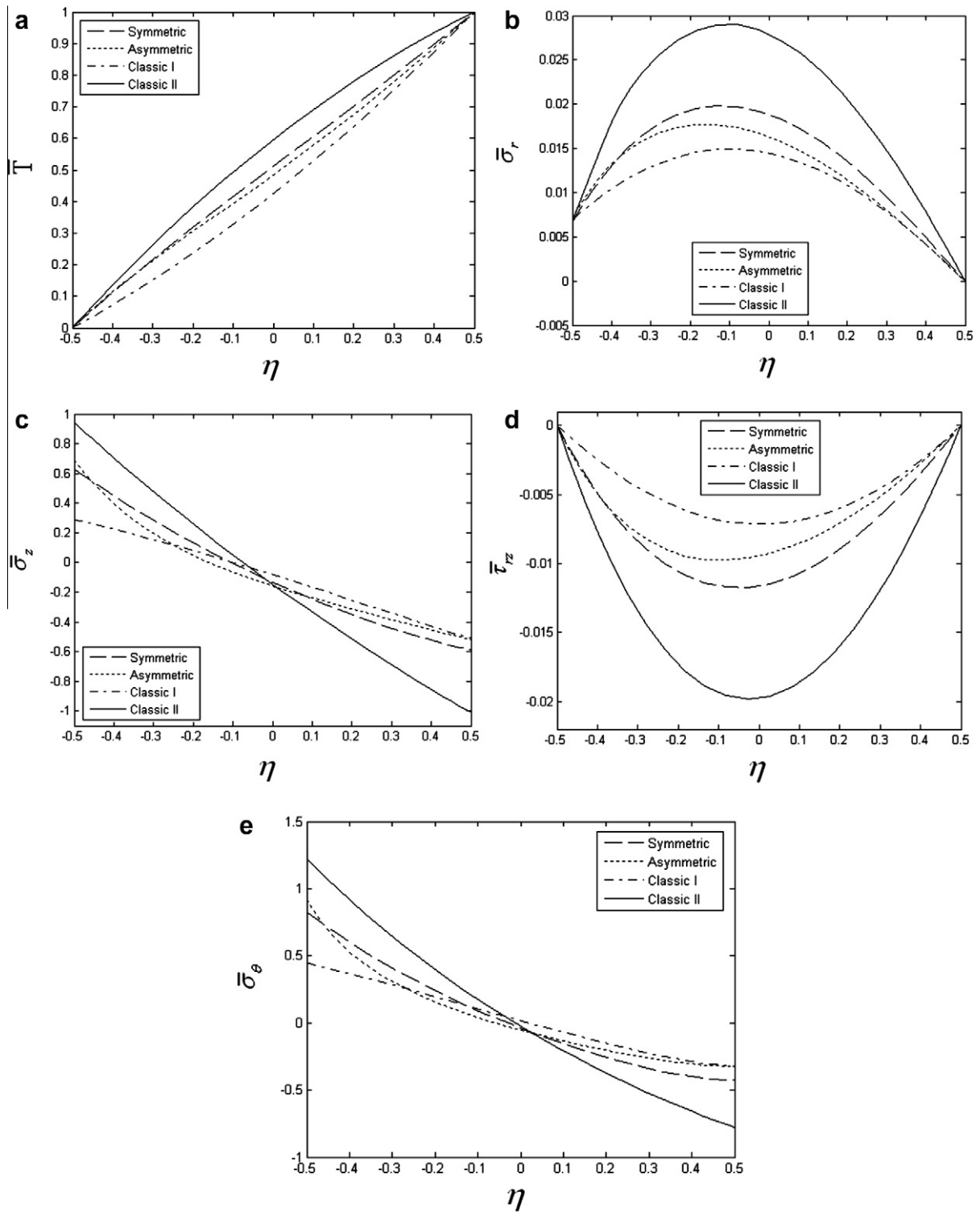


Fig. 15. Distribution of the mechanical entities along the thickness of FGM fiber-reinforced cylindrical shells resting on elastic foundation with different materials ($p = 1$, $S = 10$, $K_w = 1e2$, $K_g = 1e - 2$).

Appendix B. Expressions for the coefficients J_{ij} in Eq. (27)

$$J_{1r} = \left[-\bar{C}_{55} \left(\frac{n\pi}{L} \right)^2 - \bar{C}_{44} \frac{1}{r^2} m^2 + \frac{\partial \bar{C}_{23}}{\partial r} \frac{1}{r} - \bar{C}_{22} \frac{1}{r^2} \right] + \left[\frac{\partial \bar{C}_{33}}{\partial r} + \bar{C}_{33} \frac{1}{r} \right] \frac{\partial}{\partial r} + \bar{C}_{33} \frac{\partial^2}{\partial r^2},$$

$$J_{1\theta} = \left[\bar{C}_{44} \frac{1}{r} + \bar{C}_{22} \frac{1}{r} - \frac{\partial \bar{C}_{23}}{\partial r} \right] \frac{1}{r} m - [\bar{C}_{44} + \bar{C}_{23}] \frac{1}{r} m \frac{\partial}{\partial r},$$

$$J_{1z} = \left[-\frac{\partial \bar{C}_{13}}{\partial r} - \bar{C}_{13} \frac{1}{r} + \bar{C}_{12} \frac{1}{r} \right] \left(\frac{n\pi}{L} \right) - [\bar{C}_{13} + \bar{C}_{55}] \left(\frac{n\pi}{L} \right) \frac{\partial}{\partial r},$$

$$J_{1T} = \left[-\frac{\partial \beta_{33}}{\partial r} - \frac{1}{r} \beta_{33} + \beta_{22} \frac{1}{r} \right] - \beta_{33} \frac{\partial}{\partial r},$$

$$J_{2r} = \left[\bar{C}_{22} \frac{1}{r^2} m + \bar{C}_{44} \frac{1}{r^2} m + \frac{\partial \bar{C}_{44}}{\partial r} \frac{1}{r} m \right] + [\bar{C}_{23} + \bar{C}_{44}] \frac{1}{r} m \frac{\partial}{\partial r},$$

$$J_{2\theta} = - \left[\bar{C}_{66} \left(\frac{n\pi}{L} \right)^2 + \bar{C}_{22} \frac{1}{r^2} m^2 + \bar{C}_{44} \frac{1}{r^2} + \frac{\partial \bar{C}_{44}}{\partial r} \frac{1}{r} \right] + \left[\frac{\partial \bar{C}_{44}}{\partial r} + \bar{C}_{44} \frac{1}{r} \right] \frac{\partial}{\partial r} + \bar{C}_{44} \frac{\partial^2}{\partial r^2},$$

$$J_{2z} = -(\bar{C}_{66} + \bar{C}_{12}) \frac{1}{r} \left(\frac{n\pi}{L} \right) m,$$

$$J_{2T} = -\frac{1}{r} \beta_{22} m,$$

$$J_{3r} = \left[\bar{C}_{12} \frac{1}{r} + \bar{C}_{55} \frac{1}{r} + \frac{\partial \bar{C}_{55}}{\partial r} \right] \left(\frac{n\pi}{L} \right) + [\bar{C}_{13} + \bar{C}_{55}] \left(\frac{n\pi}{L} \right) \frac{\partial}{\partial r},$$

$$J_{3\theta} = -[\bar{C}_{12} + \bar{C}_{66}] \frac{1}{r} \left(\frac{n\pi}{L} \right) m,$$

$$J_{3z} = - \left[\bar{C}_{11} \left(\frac{n\pi}{L} \right)^2 + \bar{C}_{66} \frac{1}{r^2} m^2 \right] + \left[\frac{\partial \bar{C}_{55}}{\partial r} + \bar{C}_{55} \frac{1}{r} \right] \frac{\partial}{\partial r} + \bar{C}_{55} \frac{\partial^2}{\partial r^2},$$

$$J_{3T} = -\beta_{11} \left(\frac{n\pi}{L} \right),$$

$$J_{4T} = -k_{22} \frac{1}{r^2} m^2 - k_{11} \left(\frac{n\pi}{L} \right)^2 + k_{33} \frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial k_{33}}{\partial r} \frac{\partial}{\partial r} + k_{33} \frac{\partial^2}{\partial r^2},$$

References

- [1] M. Yamanouchi, M. Koizumi, Functionally gradient materials, in: Proceedings of the First International Symposium on Functionally Graded Materials, Functionally Graded Materials Forum, Sendai, Japan, 1990, pp. 327–332.
- [2] M. Koizumi (The concept of FGM) (Ceramic Transactions), Functionally Gradient Materials, vol. 34, 1993. pp. 3–10.
- [3] Survey for application of FGM, FGM Forum, Japan Society of Non-Traditional Technology, Dept. of Material Science and Engineering, Tsinghua Univ., Tokyo, 1991.
- [4] N. Noda, Thermal stresses in functionally graded materials, J. Therm. Stresses 22 (1999) 477–512.
- [5] T. Fuchiyama, N. Noda, Analysis of thermal stress in a plate of functionally gradient material, JSAE Rev. 16 (1995) 263–268.
- [6] B. Mirzavand, M.R. Eslami, R. Shahsiah, Effect of imperfections on thermal buckling of functionally graded cylindrical shells, AIAA J. 43 (9) (2005) 2073–2076.
- [7] B. Mirzavand, M.R. Eslami, Effect of imperfection on buckling of functionally graded cylindrical shells under axial compression, in: Proc. ISME, Isfahan, Iran, May 17–19, 2005.
- [8] B. Mirzavand, M.R. Eslami, Thermal buckling of imperfect functionally graded cylindrical shells based on the Wan–Donnell model, J. Therm. Stresses 29 (1) (2006) 37–55.
- [9] K.M. Liew, S. Kitipornchai, X.Z. Zhang, C.W. Lim, Analysis of the thermal stress behaviour of functionally graded hollow circular cylinders, Int. J. Solids Struct. (40) (2003) 2355–2380.

- [10] J.L. Pelletier, S.S. Vel, An exact solution for the steady state thermo-elastic response of functionally graded orthotropic cylindrical shells, *Int. J. Solids Struct.* 43 (2006) 1131–1158.
- [11] M. Katsuo, T. Sawa, K. Kawaguchi, H. Kawamura, Axisymmetrical thermal stress analysis of laminated composite finite hollow cylinders restricted at both ends in steady state, in: *Proceedings of the ASME International Mechanical Engineering Congress and Exposition*, 1996, 17–22.
- [12] A. Bahtui, M.R. Eslami, Coupled thermo-elasticity of functionally graded cylindrical shells, *Mech. Res. Commun.* 34 (1) (2007) 1–18.
- [13] A. Bahtui, M.R. Eslami, Generalized coupled thermoelasticity of functionally graded cylindrical shells, *Int. J. Numer. Methods Eng.* 69 (4) (2007) 676–697.
- [14] M. Jabbari, A. Bahtui, M.R. Eslami, Axisymmetric mechanical and thermal stresses in thick long FGM cylinders, *J. Therm. Stresses* 29 (7) (2006) 643–663.
- [15] Y. Obata, N. Noda, Steady thermal stresses in a hollow circular cylinder and a hollow sphere of a functionally gradient material, *J. Therm. Stresses* 17 (3) (1994) 471–487.
- [16] R. Shahsiah, M.R. Eslami, Thermal buckling of functionally graded cylindrical shell, *J. Therm. Stresses* 26 (2003) 277–294.
- [17] J.N. Reddy, Z.Q. Cheng, Three-dimensional thermo-mechanical deformations of functionally graded rectangular plates, *Eur. J. Mech. A/Solids* 20 (5) (2001) 841–855.
- [18] S.S. Vel, R.C. Batra, Three-dimensional analysis of transient thermal stresses in functionally graded plates, *Int. J. Solids Struct.* 40 (25) (2003) 7181–7196.
- [19] X. Wang, L.J. Sudak, Three-dimensional analysis of multi-layered functionally graded anisotropic cylindrical panel under thermo-mechanical loading, *Mech. Mater.* 40 (2008) 235–254.
- [20] A. Alibeigloo, Exact solution for thermo-elastic response of functionally graded rectangular plates, *Compos. Struct.* 92 (1) (2010) 113–121.
- [21] S.M. Hosseini, M. Akhlaghi, M. Shakeri, Heat conduction and heat wave propagation in functionally graded thick hollow cylinder base on coupled thermoelasticity without energy dissipation, *Heat Mass Transfer* 44 (2) (2008) 1477–1484.
- [22] B. Sobhani Aragh, M.H. Yas, Three-dimensional analysis of thermal stresses in four-parameter continuous grading fiber reinforced of cylindrical panels, *Int. J. Mech. Sci.* (52) (2010) 1047–1063.
- [23] M.H. Yas, B. Sobhani Aragh, Three-dimensional analysis for thermo-elastic response of functionally graded fiber reinforced cylindrical panel, *Compos. Struct.* 92 (2010) 2391–2399.
- [24] S. Jafari Mehrabadi, B. Sobhani Aragh, On the thermal analysis of 2-D temperature-dependent functionally graded open cylindrical shells, *Compos. Struct.* 96 (2013) 773–785.
- [25] D.N. Paliwal, H. Kanagasabapathy, K.M. Gupta, The large deflection of an orthotropic cylindrical shell on a Pasternak foundation, *Compos. Struct.* 31 (1995) 31–37.
- [26] D.N. Paliwal, R.K. Pandey, T. Nath, Free vibration of circular cylindrical shell on Winkler and Pasternak foundation, *Int. J. Press. Vessels Pip.* 69 (1996) 79–89.
- [27] R. Yang, H. Kameda, S. Takada (Shell model FEM analysis of buried pipelines under seismic loading), *Bulletin of the Disaster Prevention Research Institute*, vol. 38, Kyoto University, 1998, pp. 115–146.
- [28] J.B. Cai, W.Q. Chen, G.R. Ye, H.J. Ding, On natural frequencies of a transversely isotropic cylindrical panel on a Kerr foundation, *J. Sound Vib.* 232 (5) (2000) 997–1004.
- [29] H.T. Gunawan, T. Mikami, S. Kanie, M. Sato, Free vibration characteristics of cylindrical shells partially buried in elastic foundations, *J. Sound Vib.* 290 (2006) 785–793.
- [30] M. Farid, P. Zahedinejad, P. Malekzadeh, Three dimensional temperature dependent free vibration analysis of functionally graded material curved panels resting on two parameter elastic foundation using a hybrid semi-analytic, differential quadrature method, *Mater. Des.* 31 (2010) 2–13.
- [31] A.G. Shah, T. Mahmood, N.N. Naem, Z. Iqbal, A.H. Arshad, Vibrations of functionally graded cylindrical shells based on elastic foundations, *Acta Mech.* 211 (2010) 293–307.
- [32] T. Mori, K. Tanaka, Average stress in matrix and average elastic energy of materials with misfitting Inclusions, *Acta Metall.* 21 (1973) 571–574.
- [33] Y. Benveniste, A new approach to the application of Mori–Tanaka's theory of composite materials, *Mech. Mater.* 6 (1987) 147–157.
- [34] H. Hatta, M. Taya, Effective thermal conductivity of a misoriented short fiber composite, *J. Appl. Phys.* 58 (7) (1985) 2478–2486.
- [35] B.W. Rosen, Z. Hashin, Effective thermal expansion coefficients and specific heats of composite materials, *Int. J. Eng. Sci.* 8 (1970) 157–173.
- [36] H.-S. Shen, A comparison of buckling and post-buckling behavior of FGM plates with piezoelectric fiber reinforced composite actuators, *Compos. Struct.* 91 (2009) 375–384.
- [37] V.V. Vasiliev, E.V. Morozov, *Mechanics and analysis of composite materials*, first ed., Elsevier Science Ltd., 2001.
- [38] M.H. Yas, B. Sobhani Aragh, Elasticity solution for free vibration analysis of four-parameter functionally graded fiber orientation cylindrical panels using differential quadrature method, *Eur. J. Mech. A/Solids* 30 (2011) 631–638.
- [39] Y.C. Fung, P. Tong, *Classical and Computational Solid Mechanics*, World Scientific Publishing Co., Singapore, 2001.
- [40] Z.S. Shao, T.J. Wang, Three-dimensional solutions for the stress fields in functionally graded cylindrical panel with finite length and subjected to thermal/mechanical loads, *Int. J. Solids Struct.* 43 (2006) 3856–3874.
- [41] C. Shu, *Differential Quadrature and its Application in Engineering*, Springer, Berlin, 2000.
- [42] C. Shu, B.E. Richards, Application of generalized differential quadrature to solve two-dimensional incompressible Navier Stokes equations, *Int. J. Numer. Methods Fluids* 15 (1992) 791–798.
- [43] R. Bellman, B.G. Kashef, J. Casti, Differential quadrature: a technique for the rapid solution of nonlinear partial differential equations, *J. Comput. Phys.* 10 (1972) 40–52.
- [44] C.W. Bert, M. Malik, Differential quadrature method in computational mechanics, a review, *Appl. Mech. Rev.* (49) (1996) 1–28.
- [45] F. Tornabene, E. Viola, 2-D solution for free vibrations of parabolic shells using generalized differential quadrature method, *Eur. J. Mech. A/Solids* 27 (2008) 1001–1025.
- [46] T.K. Varadan, K. Bhaskar, Bending of laminated orthotropic cylindrical shells – an elasticity approach, *Compos. Struct.* 17 (1991) 141–156.
- [47] M. Ruhi, A. Angoshtari, R. Naghdabadi, Thermo-elastic analysis of thick-walled finite-length cylinders of functionally graded materials, *J. Therm. Stresses* 28 (2005) 391–408.
- [48] D.B. Miracle, *Aeronautical applications of metal – matrix composites*, Composite, 21, ASM International, Material Park, OH, 2001, pp. 1043–1049.